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How well can we assess the validity of non-randomized studies of medications? A systematic review of assessment tools by the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research Special Interest Group.

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Title: How well can we assess the validity of non-randomized studies of medications? A systematic review of assessment tools by the International Society for Pharmacoepidemiology Comparative Effectiveness Research Special Interest Group

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Abstract

Objective: To systematically identify assessment tools for non-randomized studies (NRS) and to determine whether they address critical elements that influence the validity of NRS findings for comparative safety and effectiveness of medications.

Design: Systematic review and Delphi survey.

Data Sources: We searched Pubmed, Embase, Google®, bibliographies of reviews and experts' advice, from inception to November 2019. In parallel, we conducted a Delphi survey among the working group of the International Society for Pharmacoepidemiology Comparative Effectiveness Research (CER) Special Interest Group (SIG) to identify key methodological challenges (domains) for NRS of medications. Tools identified through the search were evaluated against a pre-piloted framework based on the methodological domains recommended by the CER SIG.

Study selection: Checklists or scales evaluating the validity of NRS.

Data extraction: At least two authors independently extracted general information of the identified tools and content data related to the prespecified domains of the framework.

Results: Of 44 tools reviewed, 48% assess multiple NRS designs, while 27% and 23% assess respectively case-control and cohort studies only. Most tools evaluated methods for selecting study participants (98%), although none address selection bias due to depletion of susceptibles (0%). Many tools addressed the definition and measurement of exposure and outcome (86%), and measurement and control for confounders (91%). Most tools have at least one item/question on design-specific sources of bias (91%), but only a few investigate reverse causation (16%), detection bias (9%), immortal time bias (7%), lack of new-user design (5%), or active comparator design (0%). Few tools address the appropriateness of statistical analyses (33%), methods for assessing internal (35%) or external validity (26%) and statistical uncertainty in the findings (47%). None of the reviewed tools investigated all the methodological domains and subdomains.

Conclusions: The acknowledgement of fatal methodological flaws (e.g., new-user design, active comparator design, immortal time bias, depletion of susceptibles, reverse causation), and statistical

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assessment of internal and external validity is currently missing in most of the existing tools. These critical elements must be integrated to systematically investigate the validity of NRS on comparative safety and effectiveness of medications.

Systematic review protocol and registration https://osf.io/es65q

Keywords: systematic review, real world data, real world evidence, tools, non-randomized studies, cohort, case-control, comparative effectiveness research, validity.

Strengths and limitations of this study

- This is the first systematic review that investigated whether existing tools adequately assess the validity of nonrandomized studies (NRS) evaluating the comparative safety and effectiveness of medications.
- Assessment tools were identified by searching through multiple sources: relevant databases, grey literature, websites of authoritative organizations, bibliographies of previous systematic reviews and experts' suggestions.
- The framework adopted to evaluate the completeness of the tools included all the main methodological challenges suggested by worldwide experts in the field of pharmacoepidemiology and healthcare outcomes research, coming from different backgrounds (academia, industry, governmental agencies) and countries.
- Tools published in English or that could not be retrieved were omitted from this systematic review.
- The search for tools in the grey literature might not be comprehensive since it was performed through only one browser.

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Introduction

There are high expectations that real-world data (RWD), and resultant real-world evidence (RWE), will become a key source of information for the development process of pharmacological or biological therapies [1-3]. The 21st Century Cures Act and the sixth Prescription Drug User Fee Act (PDUFA VI) required the Food and Drug Administration (FDA) to explore the use of RWE and, consequently, well-designed and conducted non-randomized studies (NRS), for expediting drug approvals [4,5]. Similarly, one of the goals of the European Medicines Agency (EMA) Adaptive Pathways Initiative is to supplement clinical trial data with RWD and to eventually produce RWE as part of the approval process of new medications or indications [6].

However, the growing demand for RWD has raised concerns about the reliability of NRS to generate RWE. Due to the inherent limitations of observational analyses, the validity of NRS depends largely on the implementation of complex design and analytic methodologies. In recent reports, both FDA and EMA emphasized the need to plan and execute NRS following standards that can ensure validity and reproducibility of RWE [7,8]. Tools that assess the validity of NRS can be useful instruments for both researchers (e.g., for authors and reviewers to prevent publication of poor quality pharmacoepidemiologic research), and other stakeholders who are involved in clinical, managemental, or economic decision-making (e.g., to correctly inform guidelines and clinicians, or to guide resource allocation).

An analysis on the capability of existing tools to assess the validity of NRS of comparative safety and effectiveness of medications is currently lacking. Thus, we performed a systematic review to identify existing assessment tools for NRS and to determine whether they sufficiently address the main sources of bias that may affect the validity of NRS on the comparative safety and effectiveness of medications. This study was conducted within the Comparative Effectiveness Research Special Interest Group (CER SIG) of the International Society for Pharmacoepidemiology (ISPE) as part of the development of a framework to combine evidence from NRS with randomized clinical trials (RCT) for health care decision-making.

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Methods

The systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement [9]. Systematic review protocol and registration are available at https://osf.io/es65q.

Systematic search and eligibility criteria

We searched Pubmed and Embase from inception to November 2019 to identify existing tools that investigated the validity of NRS, specifically case-control and cohort design studies. We excluded guidelines or manuals, tools to review study protocols, tools targeting NRS of non-pharmacological interventions (e.g. surgery) or assessing only one or a few specific types of bias, and tools not available in English language. In parallel, we searched the same electronic databases for systematic reviews of assessment tools of NRS. We then extracted the references of the tools included in the systematic reviews retrieved. We also performed a general search through Google® for grey literature and reviewed any additional information from initiatives, programs or organizations, and suggestions from experts. Full details on the search strategy are reported in the Supplement (Table S1 and S2, online Supplement 1). Three reviewers (E.D., G.S., L.V.) independently removed duplicates and reviewed titles and abstracts of peer-reviewed publications or documents from the grey literature to select eligible tools. Discrepancies were resolved by consensus.

Delphi survey and prespecified framework

Concurrently, we performed a Delphi survey [10] to reach a consensus among content experts about the main methodological challenges (domains) that may threaten the validity of NRS on comparative safety and effectiveness of medications. The survey is available in the online Supplement 2. The panel of experts involved members of the SIG for CER of the ISPE. Detailed information on the Delphi methods and results is reported in the online Supplement 1.

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Domains and subdomains indicated by the Delphi respondents as major elements that can impact the validity of NRS of medications were used to develop and pilot a framework to evaluate the identified NRS tools. All domains were considered equally important. A glossary of terms used in the framework is reported in Table 1.

Data extraction

Two reviewers (E.D., L.V.) independently extracted general information of the identified tools (first author or name of the tool, year of publication or online availability of the most updated version, type of tool, scope of the tool, non-randomized study designs evaluated, number of items) and content data related to the prespecified domains of the framework. Discrepancies were resolved by consensus. We categorized the tools as checklists, defined as itemized instruments (including questionnaires) developed to identify the presence or absence of critical elements, or rating scales, defined as itemized instruments aimed to identify the performance of a study at each critical element described in the tool, using a qualitative or quantitative scale.

Data synthesis

General characteristics of the identified tools were summarized with means and standard deviations, for continuous variables, and relative frequencies, for categorical variables. The findings from the online survey and the proportion of tools assessing the prespecified elements of the framework were reported in terms of relative frequencies.

Results

Overview of Tools

Of 44 tools that met our eligibility criteria [11-45], 20 were identified through the database search of peer-reviewed literature (45%) and 24 through the general online search and other sources (55%) (Figure S1 in the online Supplement 1). General characteristics of the tools are shown in Table 2 and

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Table 3. The number of items across all tools ranged from 5 to 54, with a median of 13.5 (IQR, 10.3-22). Only three tools were designed to specifically address studies on the comparative safety and effectiveness of medications: one published in 1994 by Cho et al. [38], and the GRACE checklist and the ISPOR-AMCP-NPC tool, both published in 2014 [17,18].

Tool Formats and Scopes

Most of the tools were checklists (n = 35, 80%), and thirteen checklists included a final section to elaborate a summary judgement of the study appraisal (37%). The remaining tools were scales (n = 9, 20%), and six of them provided a section for a summary judgement (67%).

Thirty-five tools (80%) were designed as critical appraisal tools for different scopes (e.g., assessing the quality of NRS included in a systematic review, screening eligible NRS to include in systematic reviews to support clinical guidelines, supporting peer-review processes or, more general, allowing readers to interpret NRS results critically). Four tools (9%) were developed to assess the quality of reporting and were mainly intended for researchers. Five other tools (11%) combined elements of both critical appraisal and quality reporting and were for a more general audience (both researchers and readers) (Table 3).

Study Designs Addressed

Twenty-one tools (48%) were developed to assess multiple NRS designs (11 targeted cohort and case-control studies and ten others addressed also other NRS designs). Other tools specifically addressed case-control (n = 12, 27%) or cohort studies (n = 10, 23%). Ten tools (23%) were designed to assess also RCTs.

Tool Elements

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The response rate of the Delphi survey was 73% (35 respondents out of 48 participants). Detailed results are reported in the Supplement 1. Domains and subdomains indicated by the respondents as major elements that can impact the validity of NRS of medications are reported in the first column of Table 4.

Methods for selecting participants

Nearly all tools assessed methods for selecting study participants to correct selection bias (n = 43, 98%). Specifically, almost half of the tools included items related to sampling strategies (n = 19, 43%), the definition of inclusion and exclusion criteria (n = 27, 61%), and the generalizability of participants (i.e., attempts to achieve a sample of participants that represents the target population) (n = 21, 48%), while no tools explicitly addressed the depletion of susceptibles (Table 2).

Definition and measurement of exposure, outcomes, covariates, and follow-up

Forty tools had at least one item assessing the definition and measurement of exposure, outcome, covariates and follow-up (91%). Assessment of exposure and outcome was widely reported by the tools (n = 38, 86%), while definition and measurement of covariates (n = 12, 27%) or follow-up (n = 16, 36%) were less often addressed (with the exception for tools addressing follow-up in cohort studies, n = 9, 82%).

Design-specific sources of bias

Design-specific sources of bias were assessed by 91% of the tools (n = 40), and generally included loss to follow-up bias (n = 21, 48%), observer or interviewer bias (n = 11, 25%), reverse causation bias (n = 7, 16%), recall bias (n = 6, 14%), misclassification bias (n = 6, 14%), and non-contemporaneous comparator bias (n = 5, 11%). A few or no tools assessed detection or surveillance bias (n = 4, 9%), time related bias, such as immortal person-time bias or time-window bias (n = 3, 7%), and biases due to lack of new-user design (n = 2, 5%) or active comparator design (0%). Other tools reported

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only a general item/question on the risk of bias (n = 9, 20%), without any reference to a specific type of bias.

Tools specifically for cohort studies addressed more frequently loss to follow-up (n = 9, 82%) and reverse causation biases (n = 5, 45%) compared to the other tools, while tools for case-control studies addressed mostly recall (n = 4, 33%) and observer biases (n = 3, 25%). Tools for multiple NRS covered commonly loss to follow up (n = 11, 52%) and interviewer or observer biases (n = 7, 35%).

Confounding

Forty tools (91%) included at least one item or question related to confounding. Specifically, 26 tools (59%) searched whether study design was planned in a way to minimize confounding, 38 (86%) whether confounders were measured and included in the analyses, and only five whether potential unmeasured confounding was assessed in the sensitivity analyses (11%).

Appropriateness of Statistical Analyses, external and internal validity

One-third of the tools (n = 14, 32%) assessed the appropriateness of statistical analyses. Almost half (n = 21, 48%) included methods for measuring uncertainty in the findings. Few tools addressed methods for evaluating internal (n = 15, 34%) or external validity (n = 11, 25%).

These results were mostly consistent across the three different types of design addressed, cohort only, case-control only and multiple NRS, except for the assessment of follow-up (Domain 2) and several design-specific sources of biases (Domain 3) already mentioned above (Table 4). None of the reviewed tools covered all the main domains and subdomains as identified by the CER SIG and listed in Table 4. Additional results are shown in the Supplement 1.

Discussion

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In this systematic review, we identified assessment tools evaluating the validity of NRS on comparative safety and effectiveness of medications. Of 44 tools included, only three were specifically designed to assess NRS of pharmacologic interventions [17,18, 38].

Overall, we found that existing tools assessed most of the methodological challenges described by the domains of the SIG CER framework, but critical elements were often missing. For example, although many tools assessed the risk of selection bias, only half of them explicitly investigated sampling strategies and pre-specification of inclusion/exclusion criteria. Even more surprising was that no tools explored the potential for selection bias due to depletion of patients that are susceptible to the outcome. This cohort-based phenomenon occurs when new users of a medication are depleted of all susceptible subjects to the outcome, documenting an increased incidence rate of the outcome in an early stage, followed by a decreased rate with a longer duration of exposure [46]. Depletion of susceptibles is an important source of bias to account for when evaluating effects of new medications in incident users and can significantly undermine the validity of the results [46]. It should be described as source of selection bias, particularly in tools assessing NRS of comparative safety of medications.

Similarly, many tools investigated misclassification or information bias of exposure and outcome. However, only about one-third assessed definition and measurement of covariates, and less than one-fourth of the case-control and multiple-NRS designs tools assessed information on follow-up definition. Again, these are common causes of bias and should be integrated in tools that investigate the validity of NRS.

Design-specific sources of bias was a critical domain. Although overall 91% of the tools had at least one item/question investigating biases due to an inappropriate study design, only ROBINS-I and the GRACE checklist addressed bias due to lack of new user design, and time-related bias (i.e., immortal person-time bias or time-window bias), while no tools investigated bias due to lack of active comparator design. Since these biases are *fatal methodological flaws* (defined as elements that by themselves can significantly compromise the validity of the results), their assessment must be included in appraisal tools for NRS of pharmacological interventions. Recent evidence on NRS of glucose-lowering medications

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reported that only one-fourth of the studies adopted a new-user design and less than half used an active comparator [47]. Potential for time-related bias was detected in more than two-third of the studies [47]. Integrating the evaluation of these fatal methodological flaws in existing tools and recommending the use of these tools before publication can increase awareness in the clinical research community with respect to major design-specific biases. This can ultimately decrease the amount of NRS with invalid findings on the safety or effectiveness of medications.

A high percentage of tools evaluated whether confounders were appropriately measured, controlled for in the analysis, and considered in the study design. However, very few tools included at least one item/question on whether potential unmeasured confounding had been considered in the analysis or interpretation of findings.

One-third of the tools checked the appropriateness of statistical analyses, but most omitted specific reference to common flaws such as overadjustment or incorrect outcome model specification. Similarly, only one-third of the tools assessed internal validity (e.g., through sensitivity analysis to address potential confounding, measurement errors or other biases), and only one-fourth assessed external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.).

While recently published tools such as the CASP checklist, ISPOR-AMCP-NPC, RAMboMAN, GRACE and ROBINS-I [11,13,16-18] are among the most complete tools, addressing several of the critical elements underlined by the CER SIG ISPE, they all lack of items assessing two or more fatal methodological flaws (e.g., selection bias for depletion of susceptibles, immortal-time bias or window-time bias, lack of new user design, lack of active comparator design, reverse causation bias, adjustment for causal intermediaries). Assessment tools can be powerful instruments for researchers, authors, reviewers of scientific journals or readers, helping to identify the main limitations of a study and to correctly interpret the results, to acknowledge major methodological flaws, and, ultimately, to prevent publication of studies with invalid findings. Thus, it is key that available tools do assess issues that may lead to fatal methodological flaws.

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Further, other decision-makers, such as clinicians, guideline developers, and payers or investors, can benefit from instruments that help to ensure the validity of NRS findings. RCTs can be an insufficient source of evidence for decisions on pharmaceutical interventions [48, 49]. Despite well-designed and adequately powered RCTs being considered the "gold standard" of the clinical research paradigm, they can often be too time- and money-intensive. Trials are often relatively small, focus on short-term efficacy and safety in a controlled clinical environment, using surrogate outcomes, or underrepresenting high-risk populations that can be most likely the target on the new medications in the real-world setting [48, 49]. Trials might also not record treatments taken outside the study protocol [49]. Further, patients volunteered to participate in a trial are usually very motivated and so more adherent to therapy compared to the real-world population [49]. NRS of RWD can help to address these issues and could be supplemented with evidence from RCTs to provide a more complete picture on the effectiveness of pharmaceutical interventions in less controlled environments. NRS have the advantages to investigate large-scale populations, high-risk subpopulations, rare exposures, diseases or outcome, and long-term outcomes or other delayed health effects at low costs and rapidly [48, 49]. Moreover, since RWD are often collected for intents unrelated to research objectives (mainly administrative), biases such as recall bias, interviewer bias, nonresponse bias and bias for loss to follow-up are reduced or eliminated [48]. Thus, since RWE derived by NRS contribute significantly to generate evidence of comparative effectiveness research of medications, our synthesis can help numerous stakeholders to evaluate whether the NRS considered are valid enough to guide decision-making. Although checklists have been previously suggested for reviewing the risk of bias of general NRS [50], we cannot strongly recommend a specific tool for NRS on comparative analyses of medications. As already mentioned, items or questions that address all those methodological flaws must be integrated in the existing tools. Based on our findings, most recent and comprehensive tools such as ROBINS-I [16] and GRACE [18] assessed at least two design-specific fatal methodological flaws. They could therefore be prioritized in this endeavor, although ROBINS-I was not specifically designed to assess the validity of NRS on comparative safety and effectiveness of medication.

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To our knowledge this is the first systematic review that investigated whether existing tools adequately assess the validity of cohort and case-control studies evaluating the comparative safety and effectiveness of medications. Previously published systematic reviews on assessment tools for NRS were not specifically focused on pharmacological interventions [51, 52], included randomized study designs [53-55], or investigated only a specific type of bias [56]. One systematic review of NRS tools for medications focused only on safety outcomes and it is now outdated, since published in 2012 [57]. Our systematic review has multiple strengths: authors reviewed the results of the searches independently following a pre-defined protocol; the framework for data extraction was developed based on inputs of worldwide experts in the field of pharmacoepidemiology and healthcare outcomes research coming from different backgrounds (academia, industry, governmental agencies) and different countries; and it included the most updated versions of the identified tools. This review also has limitations. Search for tools in the grey literature might not be comprehensive since it was performed through only one browser. The search was also restricted to tools published in English and excluded identified tools that could not be retrieved.

In conclusion, our study has highlighted that existing tools are lacking a comprehensive assessment of major methodological flaws that can affect the validity of NRS assessing the comparative safety and effectiveness of medications. Although several well-known threats to study validity have been addressed, major aspects such as lack of new-user design, active comparator design, time-related bias (i.e., immortal time bias, time-window bias), and statistical assessment of internal validity, remain poorly addressed. Including these critical elements to existing tools may provide a more accurate instrument to evaluate NRS of pharmacological interventions to increase awareness in the clinical research community about major addressable flaws in pharmacoepidemiology. This, in turn, may further improve the validity of NRS on the comparative safety and effectiveness of medications and reduce the publication of studies with unreliable findings.

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Conflict of Interest

Authors declare no conflict of interest

Author's Contribution

All individuals listed as authors qualify for authorship according to the ICMJE guidelines. All authors conceived and designed the research. E.D. and L.V. conducted the literature search. E.P., J.M.F., G.S., T.P.A.D., E.D. and L.V. planned and drafted the Delphi survey, all other authors reviewed it. E.D. and L.V. extracted general information of the identified tools, all other authors reviewed it. E.D. performed the statistical analysis, drafted the manuscript and responded to other authors' comments. All authors reviewed, contributed to revisions and approved the final version of the manuscript.

Ethics approval

Not needed. Since the study did not involve primary data collection, the protocol was not submitted for Institutional Review Board approval.

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Transparency statement

The corresponding author affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned have been explained.

Patient and Public Involvement statement

This research was done without patient involvement. Patients were not invited to comment on the study design and were not consulted to develop patient relevant outcomes or interpret the results. Patients were not invited to contribute to the writing or editing of this document for readability or accuracy.

Data sharing

No additional data available.

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Table 1. Glossary of terms	
Term	**Definition**
Active comparator design	A study design that compares the effect of the drug of interest with another drug used in clinical practice instead of non-use.
Adjustment for causal intermediaries	Adjustment for an intermediate variable (or a descending proxy for an intermediate variable) on a causal path from exposure to outcome.
Case–control design	A study design in which cases (patients with outcomes) are identified and compared with controls (patients without outcomes) with respect to the exposure of interest.
Cohort design	A study design in which a group of patients (a cohort) is identified and followed to ascertain the occurrence of an outcome.
Confounding	A mixing of effects that arises when patients with different baseline risks are compared; the resulting effect measure is a mix of drug effects and risk factor effects.
Depletion of susceptibles	Selection bias that occurs when the initiation of exposure to a drug is associated with an early increased incidence rate of the study outcome, followed by a decreased incidence rate with longer duration of exposure (e.g., users of new drugs are compared to users of older drugs).
Detection or surveillance bias	Bias that occurs when the degree of outcome surveillance (or an associated symptom) is related to exposure and is differential among the exposure groups.
Immortal time bias	Time-related bias that derives from including a period of follow-up during which, by design, outcomes cannot occur.
Time-window bias	Time-related bias, in the context of a case-control study nested in a cohort, that derives from the use of time-windows of different lengths between cases and controls to define time-dependent exposures.
Incorrect outcome model specification	Misspecification of a statistical model that leads to biased outcome results. Common causes are omission of a relevant variable, inclusion of an unnecessary variable, adopting the wrong functional form, incorrect specification of the error term, uncertainty about what the true model is, reciprocal causation.
Loss to follow up bias	Bias that occurs when there is difference in retention during the follow-up period after enrollment that are related to exposure status and outcome.
New-user design	A study design that starts following patients at the time they initiate a new drug (also known as incident-user design)
Non-contemporaneous comparator bias	Bias generated by differences in the timing of selection of comparator group(s) within a study influence exposures and outcomes resulting in biased estimates.
Reverse causation (or reverse causality)	Bias due to direction of cause-and-effect contrary to a common presumption, or a two-way causal relationship between exposure and outcome.
Recall bias	Bias that occurs when participants do not remember previous events or experiences accurately or omit details (not for claims-based studies).
Selection bias	Bias that occurs when selection of participants or follow-up time is related to both intervention and outcome (e.g. prevalent users of a drug are compared with non-users or incident users). Our framework has a separate sub-domain that refers to selection bias due to lack of generalizability, applicability or transferability to patients who were excluded from the study.

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Table 2. General characteristics of the tools included in the systematic review

Tool identified*	Year	Type of tool	Scope of the tool	Study design evaluated	Items
RELEVANT	2019	Checklist	Critical appraisal and reporting	NRS	21
RAMboMAN - GATE-EPIQ	2019	Rating scale + summary judgment	Critical appraisal	Coh (+RCTs), CC	Coh (+RCTs) 21, CC 18
MMAT	2018	Checklist	Critical appraisal	NRS	5
CASP	2018	Checklist	Critical appraisal	Coh, CC	Coh 12, CC 11
SURE	2018	Checklist + summary judgment	Critical appraisal	Coh, CC	Coh 13, CC 11
JBI	2017	Checklist + summary judgment	Critical appraisal	Coh, CC	Coh 11, CC 10
ROBINS-I	2016	Checklist + summary judgment	Critical appraisal	NRS	34 (+ 8 optional question)
ISPOR-AMCP-NPC (#)	2014	Checklist + summary judgment	Critical appraisal	Coh CC	32
GRACE (#)	2014	Checklist + summary judgment	Critical appraisal	Coh CC	11
NIH–NHLBI	2014	Checklist + summary judgment	Critical appraisal	Coh (+ CSS), CC	Coh (+ CSS) 14, CC 12
HEBW	2014	Checklist + summary judgment	Critical appraisal	Coh	18
RoBANS	2013	Rating scale	Critical appraisal	NRS	6
RTI-Item Bank	2013	Checklist	Critical appraisal	NRS	13
Newcastle-Ottawa	2013	Rating scale + summary judgment	Critical appraisal	Coh, CC	Coh 8, CC 8
SIGN - Version 3.0	2012	Checklist + summary judgment	Critical appraisal	Coh, CC	Coh 14, CC 11
Montreal	2011	Checklist	Critical appraisal	Coh CC (+RCTs)	10
EPHPP	2011	Rating scale	Critical appraisal	Coh CC (+RCTs)	17
STROBE - Version 4	2007	Checklist	Reporting	Coh, CC	Coh 22, CC 22
TREND	2004	Checklist	Reporting	NRS	22
Margetts	2002	Checklist	Reporting	Coh CC	11
Zaza	2000	Checklist	Critical appraisal	Coh CC	15
Downs-Black	1998	Rating scale	Critical appraisal and reporting	Coh CC (+RCTs)	27
Elwood	1998	Checklist	Critical appraisal	Coh CC (+RCTs)	20
Hadorn	1996	Checklist	Critical appraisal	Coh (+RCTs)	7
London	1996	Checklist	Critical appraisal	Coh CC	33
Avis	1994	Rating scale + summary judgment	Critical appraisal and reporting	Coh CC (+RCTs)	24
Durant	1994	Checklist	Critical appraisal	CC	23
Levine	1994	Checklist	Critical appraisal	Coh CC (+RCTs)	10
Gyorkos	1994	Checklist	Critical appraisal	Coh, CC	Coh 6, CC 5
Cho (#)	1994	Rating scale + summary judgment	Critical appraisal	NRS (+RCTs)	24
COEH	1991	Checklist	Critical appraisal	NRS	54
Fowkes-Fulton	1991	Checklist + summary judgment	Critical appraisal and reporting	NRS (+RCTs)	6
Lichtenstein	1987	Checklist	Critical appraisal	CC	20
Gardner	1986	Checklist	Critical appraisal	NRS	12
Horwitz	1979	Checklist	Critical appraisal and reporting	CC	12

Notes: Nine records from our bibliographic search provided two separate tools to assess cohort or case-control studies. Thus, the overall number of included records is 34, while the number of included assessment tools is 43.
NRS = non-randomized study, Coh = cohort study, CC = case control, RCT = randomized controlled trial, CSS = cross-sectional study, NRS = Non-randomized studies.
* tool name or first author name, if the tool does not have an assigned name and it was published in peer-review journals.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.
Acronym of the tools: RELEVANT = The REal Life EVidence AssessmeNt Tool; RAMboMAN, GATE-EPIQ = Recruitment Allocation Maintenance blind objective Measurements Analyses, Graphic Approach To Epidemiology - Effective Practice, Informatics and Quality Improvement; MMAT = Mixed Methods Appraisal Tool; CASP = The Critical Appraisals Skills Programme; SURE Specialist Unit for Review Evidence; JBI = The Joanna Briggs Institute; ROBINS-I = Risk Of Bias In Non-randomized Studies - of Interventions; ISPOR-AMCP-NPC =

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International Society for Pharmacoeconomics and Outcomes Research - Academy of Managed Care Pharmacy - National Pharmaceutical Council; GRACE = The Good ReseArch for Comparative Effectiveness; NIH–NHLBI = The National Institute of Health - The National Heart, Lung, and Blood Institute; HEBW = Health Evidence Bulletins Wales; RoBANS = Risk of Bias Assessment tool for Non-randomized Studies; RTI-Item Bank = Research Triangle Institute Item Bank; SIGN = The Scottish Intercollegiate Guidelines Network; STROBE = STrengthening the Reporting of OBservational studies in Epidemiology; EPHPP = Effective Public Health Practice Project Quality Assessment Tool; TREND = Transparent Reporting of Evaluations with Nonrandomized designs; COEH = Centre for Occupational and Environmental Health of The University of Manchester.

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Table 3. Methodological challenges addressed by the included tools.				
Characteristics	**All, n = 44**	**Cohort *, N = 11**	**Case-control, n = 12**	**NRS**, n = 21**
Publication year – n (%)				
1979-1989	3 (7)	0 (0)	2 (17)	1 (5)
1990-1999	12 (27)	2 (18)	2 (17)	8 (38)
2000-2009	5 (11)	1 (9)	1 (8)	3 (14)
2010-2019	24 (55)	8 (73)	7 (58)	9 (43)
Type of tool – n (%)				
Checklist	22 (50)	4 (36)	6 (50)	12 (57)
Checklist + summary judgment	13 (30)	5 (45)	4 (33)	4 (19)
Rating scale	3 (7)	0 (0)	0 (0)	3 (14)
Rating scale + summary judgment	6 (14)	2 (18)	2 (16)	2 (9)
Scope of the tool – n (%)				
Critical appraisal	35 (80)	9 (81)	10 (83)	16 (76)
Reporting	4 (9)	2 (18)	1 (8)	1 (5)
Critical appraisal and Reporting	5 (11)	0 (0)	1 (8)	4 (19)
Tools designed for CER – n (%)	3 (7)	0 (0)	0 (0)	3 (14)
Number of items – median (IQR)	13 (10.3-21.8)	13 (9.5-16)	11.5 (10.8-18.5)	17 (11-24)

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Table 4. Methodological challenges addressed by the included tools.

Domains		Cohort tools*, N = 11	Case-control tools, n = 12	NRS tools**, n = 21	Total, n = 44
1. Methods for selecting participants – n (%)		11 (100)	12 (100)	20 (95)	43 (98)
	Sampling strategies to correct selection bias	4 (36)	6 (50)	9 (42)	19 (43)
	Inclusion and exclusion criteria of target population	6 (55)	8 (67)	13 (61)	27 (61)
	Depletion of susceptibles	0 (0)	0 (0)	0 (0)	0 (0)
	External validity of target population	6 (55)	6 (50)	9 (43)	21 (48)
	Others***	11 (100)	12 (100)	18 (86)	41 (93)
2. Definition and measurement of exposure, outcomes, covariates, and follow-up – n (%)		11 (100)	12 (100)	17 (81)	40 (91)
	Definition and measurement of exposure	11 (100)	11 (92)	16 (76)	38 (86)
	Definition and measurement of outcomes	11 (100)	11 (92)	16 (76)	38 (86)
	Definition and measurement of covariates	4 (36)	4 (33)	4 (19)	12 (27)
	Definition and measurement of follow-up	9 (82)	3 (25)	4 (19)	16 (36)
3. Design-specific sources of bias – n (%)		11 (100)	10 (83)	19 (90)	40 (91)
	New user design	0 (0)	0 (0)	2 (10)	2 (5)
	Active comparator design	0 (0)	0 (0)	0 (0)	0 (0)
	Immortal time bias or time-window bias	0 (0)	0 (0)	3 (14)	3 (7)
	Detection or surveillance bias	1 (9)	2 (17)	1 (5)	4 (9)
	Loss to follow-up bias	9 (82)	1 (8)	11 (52)	21 (48)
	Non-contemporaneous comparator bias	0 (0)	1 (8)	4 (19)	5 (11)
	Reverse causation	5 (45)	1 (8)	1 (5)	7 (16)
	Misclassification bias	1 (9)	2 (17)	3 (15)	6 (14)
	Recall bias^	1 (9)	4 (33)	1 (5)	6 (14)
	Interviewer or observer bias^	1 (9)	3 (25)	7 (35)	11 (25)
	Ascertainment bias^	0 (0)	1 (8)	1 (5)	2 (5)
	General item/question on bias^	3 (27)	3 (25)	3 (14)	9 (20)
	Other biases^^	0 (0)	2 (17)	5 (24)	7 (16)
4. Confounding – n (%)		11 (100)	11 (92)	18 (86)	40 (91)
	Study design used to minimize confounding	6 (55)	7 (58)	13 (62)	26 (59)
	Confounders measured and included in statistical analyses	10 (91)	10 (83)	18 (86)	38 (86)
	Potential unmeasured confounding addressed in the analysis (e.g., proxy analysis, IV analysis)	1 (9)	1 (8)	3 (14)	5 (11)
5. Lack of appropriateness of statistical analyses (with specific mention of overadjustment, and/or incorrect outcome model specification) – n (%)		2 (18)	3 (25)	9 (43)	14 (32)
6. Methods for assessing statistical uncertainty in the findings (e.g., confidence intervals reported for each analysis) – n (%)		7 (64)	6 (50)	8 (38)	21 (48)
7. Methods for assessing internal validity (e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases) – n (%)		3 (27)	3 (25)	9 (43)	15 (34)
8. Methods for assessing external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.) – n (%)		4 (36)	3 (25)	4 (19)	11 (25)

* two tools evaluated both cohort and RCTs together, one tool evaluated both cohort and cross-sectional studies together;
** NRS tools refer to a single tool built to evaluate both cohort and case-control studies, or a tool built to evaluate additional NRS (e.g. cross-sectional studies, before-after studies) together with cohort and case-control studies. Eight NRS tools included also the evaluation of RCTs;
*** "others" refers to items not included in our evaluation framework but included in the reviewed tools to investigate selection bias (e.g., population characteristics sufficiently described to determine the applicability of the research question, sample size justification and power description, ethical considerations etc.)
^ design-specific biases not included in the evaluation framework but addressed by the reviewed tools
^^ other design-specific biases not included in the evaluation framework but addressed by a few tools (e.g., bias due to missing data, patients' blinding, different length of follow-up between groups, Berkson's bias, protopathic bias) ^

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Supplement file 1

Table S1. Search strategy

Table S2. List of investigated websites of authoritative initiatives, programs or organizations.

Figure S1. Flowchart of the Delphi procedure

Figure S2. Flowchart of the selection of the tools

Table S3. Excluded tools after full-text screening

Figure S3. Assessment tools that included or not at least one question/item to address subdomains of methods for selecting study participants (Domain 1), stratified by study design.

Figure S4. Assessment tools that included or not at least one question/item to address subdomains for definition and measurement of exposure, outcomes, covariate variables and follow-up (Domain 2), stratified by study design.

Figure S5. Assessment tools that included or not at least one question/item to address subdomains to detect design-specific sources of biases (Domain 3), stratified by study design.

Figure S6. Assessment tools that included or not at least one question/item to address confounding (Domain 4), stratified by study design**.**

Figure S7. Assessment tools that included or not at least one question/item to address lack of appropriateness of statistical analyses (Domain 5), methods for assessing statistical uncertainty in the findings (Domain 6), methods for assessing internal validity (Domain 7), methods for assessing external validity (Domain 8), stratified by study design.

Figure S8. Distribution of domains for each reviewed tool

Figure S9. Distribution of domains for each reviewed tool of case-control studies only

Figure S10. Distribution of domains for each reviewed tool of cohort studies only

Figure S11. Distribution of domains for each reviewed tool of multiple study designs (NRS)

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In this supplement, we describe in greater detail the methods and results of our review process and Delphi survey and report additional findings related to the evaluation of each assessment tool.

As stated in the main manuscript, the framework to evaluate the assessment tools was developed based on the domains and subdomains indicated by the CER SIG as the main critical elements that can threat to the validity of NRS of medications.

To collect the domains and to ultimately develop the evaluation framework, we performed a survey following the Delphi methodology. This is an iterative multistage process that uses rounds of questionnaires to reach a consensus among content experts [Hasson et al. 2000]. The survey targeted all members of the Special Interest Group (CER) of the International Society for Pharmacoepidemiology Comparative Effectiveness Research (SIG). These experts were informed about the survey during an online meeting hold in July 2019. During the meeting, study protocol and Delphi survey were presented to the group, addressing that answering to the survey would have constituted consent to participate in the study. After the online meeting, the study protocol was e-mailed to the members of CER SIG.

On August 2019, a link to a self-administered online questionnaire was sent by email to 48 CER SIG members. In the first page of the survey, a cover letter outlined the purpose of the study and assured the participants of anonymity in processing the results. The online questionnaire could have only been submitted once per person. The experts had three weeks to answer after the initial e-mail, during this time one remainder was sent.

The two-section questionnaire was developed through several face-to-face and online meetings among the authors. The first section included questions on demographic and professional information of participants. In the second section, respondents were asked to rate, on a five-point scale ranging from 1 (“not important”) to 5 (“extremely important”), which items should be considered when assessing the overall quality of a NRS, specifically cohort and case-control studies, of pharmacological interventions. In addition to the rating, the respondents had the option of providing any other relevant considerations at the end of the questionnaire.

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The outcome variable of the online survey, originally consisting of five categories, was collapsed into two levels. Participants considered important an item if they rated it 4 or 5 (where 5 is "extremely important"), while they did not consider an item of key importance if they rated it 1, 2 or 3 (where 1 is "not important"). When at least 70% of participants agreed on the importance of an item (rating 1, 2 or 3 vs rating 4 or 5), we considered that a consensus had been reached on that item. After participants ranked the relevance of the items and provided comments to the first questionnaire, we reviewed the responses and adjusted the survey to reflect the feedback. Based on experts' suggestions, we added: a subdomain to domain 1 "Methods of selecting participants" to address the external validity or generalizability of the study population to the eligible or target population; the subdomain "Misclassification bias" to domain 3 "Design-specific sources of bias". We removed "Recall bias" from the critical elements of domain 3. The reason was that since RWD are often collected for intents unrelated to pursue research goals (e.g., administrative purposes), biases such as recall bias, interviewer bias, nonresponse bias are usually reduced or even eliminated [Sørensen HT et al.]. We still collected information on other design-specific biases not included in the current framework.

A revised survey was returned to the same participants for reassessment of the content. Since in this second round the threshold of 70% was reached for all the old and new items, we terminated the consensus procedure and communicated the results to the CER SIG members. The Delphi flowchart is shown in Figure S1, while the last version of cover letter and online questionnaire is available on the online Supplement 2.

The evaluation framework was built on the main domains and subdomains identified through the Delphi process and it was used as an instrument to evaluate the completeness of the existing assessment tools for cohort and case-control studies only or multiple NRS. The assessment of each domain and subdomain and the distribution of domains in the reviewed tools are shown in Figures S3-S11.

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Table S1. Search strategy

Items	N.	Terms
Critical appraisal tool	#1	"critical" [All Fields] AND "appraisal" [All Fields] AND "tools" [All Fields]
	#2	"critical" [All Fields] AND "appraisal" [All Fields]
	#3	("critical" [All Fields] AND "review" [All Fields]) OR "critical review" [All Fields]) AND form [All Fields]
	#4	("systematic review" [Publication Type] OR "systematic reviews as topic" [MeSH Terms] OR "systematic review"[All Fields]) AND form [All Fields]
	#5	appraisal [All Fields] AND ("research design" [MeSH Terms] OR ("research" [All Fields] AND "design" [All Fields]) OR "research design" [All Fields] OR ("research" [All Fields] AND "methodology" [All Fields]) OR "research methodology"[All Fields])
	#6	("research design" [MeSH Terms] OR ("research" [All Fields] AND "design" [All Fields]) OR "research design" [All Fields]) AND ("review" [Publication Type] OR "review literature as topic" [MeSH Terms] OR "review"[All Fields])
Study reporting tool	#7	"study" [All Fields] AND "reporting" [All Fields] AND "tool" [All Fields]
	#8	"study" [All Fields] AND "reporting" [All Fields]
	#9	"reporting" [All Fields] AND "form" [All Fields] AND ("Studies"[Journal] OR "studies"[All Fields])
	#10	"reporting" [All Fields] AND ("Studies"[Journal] OR "studies"[All Fields])
Tool	#11	"checklist" [MeSH Major Topic] OR "scale*" [Title/Abstract]
	#12	"surveys and questionnaires"[MeSH Major Topic] OR "questionnaire*" [Title/Abstract]
	#13	("tool*" [All Fields] OR "instrument*" [All Fields] OR "checklist*" [All Fields] OR "questionnaire*" [All Fields]) AND ("quality" [All Fields] OR "method*" [All Fields] OR "bias" [All Fields])
Study design	#14	"cohort studies"[MeSH Terms] OR cohort studies [Text Word] OR cohort stud* [All Fields]
	#15	"case-control studies" [MeSH Terms] OR case-control studies [Text Word] OR case control stud* [All Fields]
	#16	Non [All Fields] AND ("random allocation"[MeSH Terms] OR randomized [Text Word]) AND stud* [All Fields]
Systematic review	#17	"systematic review" [Publication Type] OR "systematic reviews as topic"[MeSH Terms] OR "systematic review"[All Fields]
Filters	#18	"humans"[MeSH Terms]
	#19	"Review" [ptyp] OR "systematic" [sb]
Strings		
1st search - tools*	#20	(#1 OR #2 OR #3 OR #4 OR #5 OR #6) AND (OR #14 OR #15 OR #16) AND #18
	#21	(#7 OR #8 OR #9 OR #10) AND OR #14 OR #15 OR #16) AND #18
	#22	(#11 OR #12 OR #13) AND OR #14 OR #15 OR #16) AND #18
2nd search - systematic reviews of tools*	#23	(#1 OR #2 OR #3 OR #4 OR #5 OR #6) AND (OR #14 OR #15 OR #16) AND #18 AND (#17 OR #19)
	#24	(#7 OR #8 OR #9 OR #10) AND OR #14 OR #15 OR #16) AND #18 AND (#17 OR #19)
	#25	(#11 OR #12 OR #13) AND OR #14 OR #15 OR #16) AND #18 AND (#17 OR #19)

*tools are defined as instruments (e.g., qualitative checklists, questionnaires, scoring scales, etc.) that investigate the overall quality of a study, identifying potential biases – either used to critically appraise studies included in a systematic review, or to help in the peer-reviewing process of scientific publications – (i.e., critical appraisal tools), or that support the reporting of research methods and findings (i.e. study reporting tools).
Note: the strings were built in Medline and then adapted to Embase (through Elsevier)

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Table S2. List of investigated websites of initiative, programs, and organizations

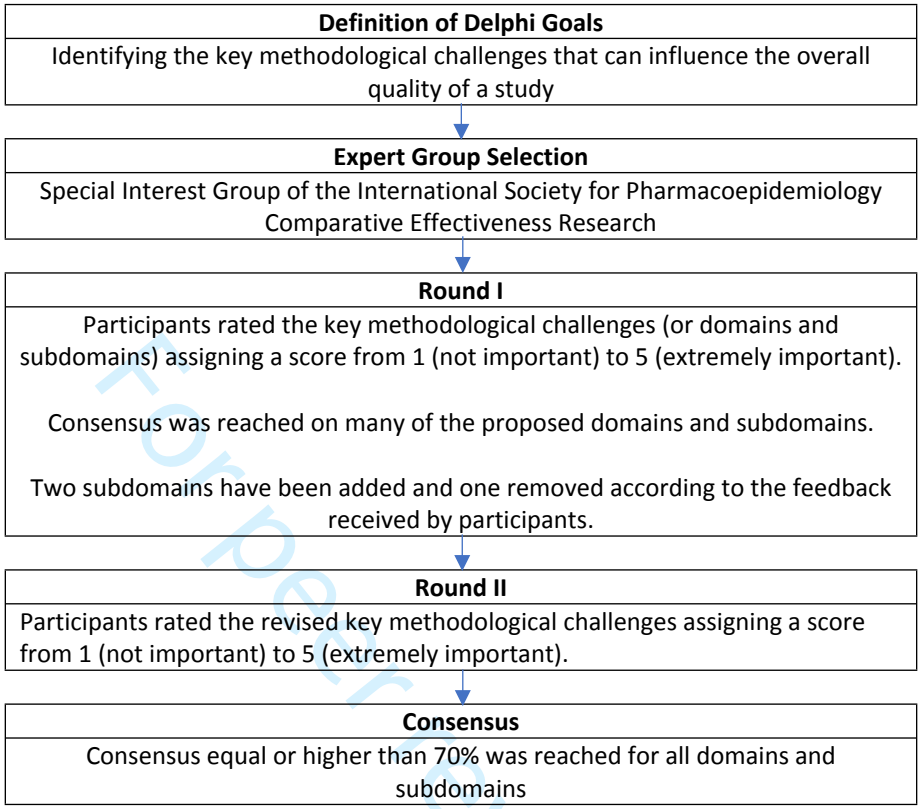
Name of initiative, programs or organization	Link to the website
STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) statement	https://strobe-statement.org/index.php?id=available-checklists
Effective Health Care (EHC) Program of the Agency for Healthcare Research and Quality (AHRQ)	https://effectivehealthcare.ahrq.gov/research-methods
The Ottawa Hospital – Research Institute	http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp
The Scottish Intercollegiate Guidelines Network (SIGN)	https://www.sign.ac.uk/checklists-and-notes.html
GRACE (Good ReseArch for Comparative Effectiveness) initiative	https://www.graceprinciples.org/about-grace.html
The Joanna Briggs Institute	http://joannabriggs.org/research/critical-appraisal-tools.html
TREND (Transparent Reporting of Evaluations with Nonrandomized Designs) statement	https://www.cdc.gov/trendstatement/index.html
CASP (Critical Appraisal Skills Programme)	https://casp-uk.net/casp-tools-checklists/

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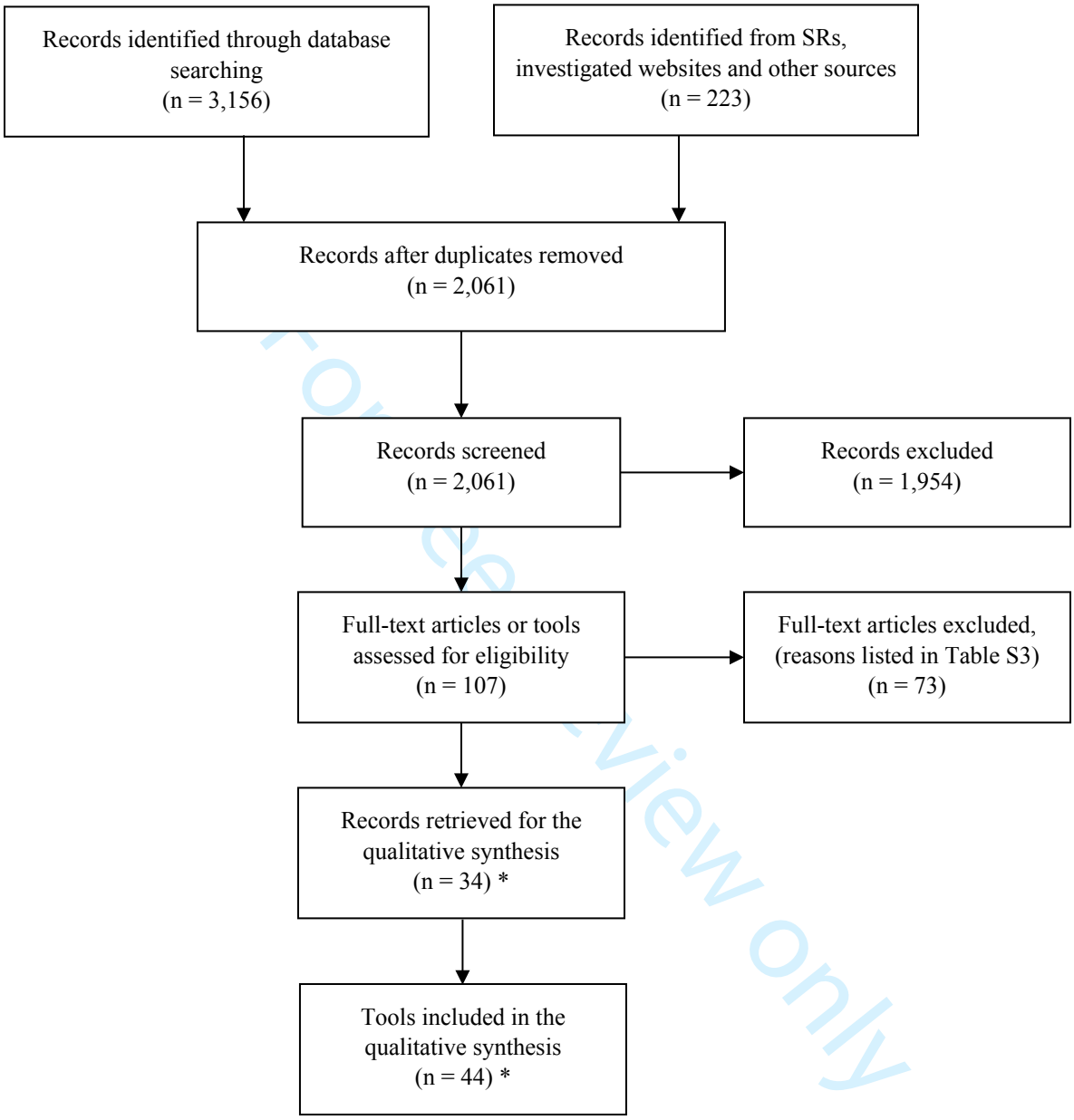
Figure S1. Flowchart of the Delphi procedure.



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Figure S2. Flowchart of the selection of the tools.



*Ten retrieved articles/records included two different assessment tools for case-control and cohort studies. Thus, the overall number of included tools raised up to 44.

The systematic reviews (SRs) were retrieved through a systematic search. Additional details about the SRs search are available upon request.

NB. The reporting items in the flowchart follows the indications given by the PRISMA Statement at Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097.

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Table S3. Excluded tools after full-text screening (73)

Reference	Reason for exclusion
Berkman ND, Lohr KN, Ansari M, et al. Chapter 15 Appendix A: A Tool for Evaluating the Risk of Reporting Bias (in Chapter 15: Grading the Strength of a Body of Evidence When Assessing Health Care Interventions for the Effective Health Care Program of the Agency for Healthcare Research and Quality: An Update). Methods Guide for Comparative Effectiveness Reviews (Prepared by the RTI-UNC Evidence-based Practice Center under Contract No. 290-2007-10056-I). AHRQ Publication No. 13(14)-EHC130-EF. Rockville, MD: Agency for Healthcare Research and Quality, 2013	Assessment of systematic reviews
Balshem H, Stevens A, Ansari M, et al. Finding grey literature evidence and assessing for outcome and analysis reporting biases when comparing medical interventions: AHRQ and the Effective Health Care Program. (Prepared by the Oregon Health and Science University and the University of Ottawa Evidence-based Practice Centers under Contract Nos. 290-2007-10057-I and 290-2007-10059-I.) AHRQ Publication No. 13(14)-EHC096-EF. Rockville, MD: Agency for Healthcare Research and Quality, 2013	Assessment of reporting bias, guidance
Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 2008; 336:924–6.	Assessment to provide strength of recomm.
Hayden JA van der Windt DA, Cartwright JL, et al. Assessing bias in studies of prognostic factors. Ann Intern Med 2013; 158:280–6.	Assessment of studies of prognostic factors
Higgins JPT, Altman DG, Sterne JAC. Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S, eds. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. London: The Cochrane Collaboration, 2011.	Assessment of RCTs
Higgins JPT, Savović J, Page MJ, et al. Revised Cochrane risk of bias tool for randomized trials (RoB 2.0), Version 20. 2016 Available at: https://sites. google.com/site/riskofbiastool/ (accessed 19 Sep 2019)	Assessment of RCTs
Hooijmans CR, Rovers MM, de Vries RB, et al. SYRCLE's risk of bias tool for animal studies. BMC Med Res Methodol 2014; 14:43	Assessment of animal studies
Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (MINORS): development and validation of a new instrument. ANZ J Surg. 2003;73(9): 712-716.	Assessment of NRS of surgical interventions
Centre for reviews and dissemination. Systematic Reviews: CRD's Guidance for Undertaking Reviews in Healthcare. 3rd ed. UK: University of York; 2009.	No assessment tool, guidance for reviews
Deeks JJ, Dinnes J, D'Amico R, et al. Evaluating non-randomised intervention studies. Health Technol. Assess (Winchester, England. 2003;7(27): iii-x):1-173.	No assessment tool, original research
NICE. Guide to the methods of technology appraisal 2013. Updated 2016.	No assessment tool, guidance for HTA
University of Wales College of Medicine. Critical Appraisal Forms. University of Wales, 2004.	N/A
Loney PL, Chambers LW, Bennett KJ, Roberts JG, Stratford PW. Critical appraisal of the health research literature: prevalence or incidence of a health problem. Chronic Dis Canada 2000;19:170–77.	No structured assessment tool
Cameron I, Crotty M, Currie C et al. Geriatric rehabilitation following fractures in older people: a systematic review. Health Technol Assess 2000;4: i–111.	No assessment tool
Carneiro AV. Critical appraisal of prognostic evidence: practical rules. Rev Port Cardiol 2002; 21:891–900.	Assesment tool of prognostic stiudies
Centre for Evidence-Based Mental Health, University of Oxford. Critical Appraisal Forms. 2004. Available at: http://www.cebmh.com/ (accessed 19 Sep 2019)	Assessment tool tailored to a particular disease/topic

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Esdaile JM, Horwitz RI. Observational studies of cause-effect relationships: an analysis of methodologic problems as illustrated by the conflicting data for the role of oral contraceptives in the etiology of rheumatoid arthritis. J Chronic Dis 1986; 39:841–52.	No structured assessment tool
Khan KS, Riet GT, Popay J, Nixon J, Kleijnen J. Undertaking systematic reviews of research effectiveness. CRD's guidance for those carrying out or commissioning reviews. CRD Report number 4 (2nd edn). 2001. The University of York Centre for Reviews and Dissemination.	No structured assessment tool
How to read clinical journals: IV. To determine etiology or causation. Can Med Assoc J. 1981;124(8):985-990.	No structured assessment tool for NRS
Mulrow CD, Lichtenstein MJ. Blood glucose and diabetic retinopathy: a critical appraisal of new evidence. J Gen Intern Med 1986; 1:73–77.	Assessment tool tailored to a particular disease/topic
Whiting P, Rutjes AW, Reitsma JB, Bossuyt PM, Kleijnen J. The development of QUADAS: a tool for the quality assessment of studies of diagnostic accuracy included in systematic reviews. BMC Med Res Methodol 2003; 3:25.	Assessment tool for diagnostic accuracy studies
Campbell H, Rudan I. Interpretation of genetic association studies incomplex disease. Pharmacogenomics J 2002; 2:349–60.	Assessment tool tailored to a particular disease/topic
Solomon DH, Bates DW, Panush RS, Katz JN. Costs, outcomes, and patient satisfaction by provider type for patients with rheumatic and musculoskeletal conditions: a critical review of the literature and proposed methodologic standards. Ann Intern Med 1997; 127:52–6	Assessment tool tailored to a particular disease/topic
The STARD Group. The STARD Initiative—Towards Complete and Accurate Reporting of Studies on Diagnostic Accuracy. http://www.consort-statement.org/stardstatement.htm, 2001.	Assessment tool for diagnostic accuracy studies
Critical appraisal: Guidelines for the critical appraisal of a paper. Surgical-Tutor.org.uk, 2004.	N/A
Zola P, Volpe T, Castelli G et al. Is the published literature a reliable guide for deciding between alternative treatments for patients with early cervical cancer? Int J Radiat Oncol Biol Phys 1989;16:785–97.	No assessment tool, original research
Bollini P, Garcia Rodriguez LA, Gutthann SP, Walker AM. The impact of research quality and study design on epidemiologic estimates of the effect of nonsteroidal anti-inflammatory drugs on upper gastrointestinal tract disease. Arch Intern Med 1992; 152:1289–95.	No assessment tool, original research
Ciliska D, Hayward S, Thomas H et al. A systematic overview of the effectiveness of home visiting as a delivery strategy for public health nursing interventions. Can J Public Health 1996;87:193–98.	No assessment tool, original research
Cowley DE. Prostheses for primary total hip replacement. A critical appraisal of the literature. Int J Technol Assess Health Care 1995;11:770–78.	No assessment tool, original research
Spitzer WO, Lawrence V, Dales R et al. Links between passive smoking and disease: a best-evidence synthesis. A report of the Working Group on Passive Smoking. Clin Invest Med 1990;13: 17–42.	No assessment tool, original research
Steinberg EP, Eknoyan G, Levin NW et al. Methods used to evaluate the quality of evidence underlying the National Kidney Foundation- Dialysis Outcomes Quality Initiative Clinical Practice Guidelines: description, findings, and implications. Am J Kidney Dis 2000; 36:1–11.	No assessment tool, original research
Anders JF, Jacobson RM, Poland GA, Jacobsen SJ, Wollan PC. Secondary failure rates of measles vaccines: a meta-analysis of published studies. Pediatr Infect Dis J 1996; 15:62–66.	No assessment tool, original research
Ariens GA, van Mechelen W, Bongers PM, Bouter LM, van der WG. Physical risk factors for neck pain. Scand J Work Environ Health 2000; 26:7–19.	No assessment tool, original research
Berlin JA, Colditz GA. A meta-analysis of physical activity in the prevention of coronary heart disease. Am J Epidemiol 1990; 132:612–28.	No assessment tool, original research
Bhutta AT, Cleves MA, Casey PH, Cradock MM, Anand KJS. Cognitive and behavioral outcomes of school-aged children who were born preterm: a meta-analysis. J Am Med Assoc 2002;288: 728–37.	No assessment tool, original research

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Borghouts JA, Koes BW, Bouter LM. The clinical course and prognostic factors of non-specific neck pain: a systematic review. Pain 1998; 77:1–13.	No assessment tool, original research
Campos-Outcalt D, Senf J, Watkins AJ, Bastacky S. The effects of medical school curricula, faculty role models, and biomedical research support on choice of generalist physician careers: a review and quality assessment of the literature. Acad Med 1995; 70:611–19.	No assessment tool, original research
Carson CA, Fine MJ, Smith MA, Weissfeld LA, Huber JT, Kapoor WN. Quality of published reports of the prognosis of community-acquired pneumonia. J Gen Intern Med 1994; 9:13–19.	No assessment tool, original research
Corrao G, Bagnardi V, Zambon A, Arico S. Exploring the doseresponse relationship between alcohol consumption and the risk of several alcohol-related conditions: a meta-analysis. Addiction 1999; 94:1551–73.	No assessment tool, original research
Garber BG, Hebert PC, Yelle JD, Hodder RV, McGowan J. Adult respiratory distress syndrome: a systemic overview of incidence and risk factors. Crit Care Med 1996; 24:687–95.	No assessment tool, original research
Goodman SN, Berlin J, Fletcher SW, Fletcher RH. Manuscript quality before and after peer review and editing at Annals of Internal Medicine. Ann Intern Med 1994; 121:11–21.	No assessment tool
Jabbour M, Osmond MH, Klassen TP. Life support courses: are they effective? Ann Emerg Med 1996; 28:690–98.	No assessment tool, original research
Kreulen CM, Creugers NH, Meijering AC. Meta-analysis of anterior veneer restorations in clinical studies. J Dent 1998; 26:345–53.	No assessment tool, original research
Krogh CL. A checklist system for critical review of medical literature. Med Educ 1985; 19:392–95.	No assessment tool for NRS validity
Littenberg B, Weinstein LP, McCarren M et al. Closed fractures of the tibial shaft. A meta-analysis of three methods of treatment. J Bone Joint Surg Am 1998; 80:174–83.	No assessment tool, original research
Longnecker MP, Berlin JA, Orza MJ, Chalmers TC. A meta-analysis of alcohol consumption in relation to risk of breast cancer. JAMA 1988; 260:652–56.	No assessment tool, original research
Macfarlane TV, Glenny AM, Worthington HV. Systematic review of population-based epidemiological studies of oro-facial pain. J Dent 2001;29:451–67.	No assessment tool, original research
Manchikanti L, Singh V, Vilims BD, Hansen HC, Schultz DM, Kloth DS. Medial branch neurotomy in management of chronic spinal pain: systematic review of the evidence. Pain Physician 2002;5:405–18.	No assessment tool, original research
Meijer R, Ihnenfeldt DS, van Limbeek J, Vermeulen M, de Haan RJ. Prognostic factors in the subacute phase after stroke for the future residence after six months to one year. A systematic review of the literature. Clin Rehabil 2003;17:512–20.	No assessment tool, original research
Nguyen QV, Bezemer PD, Habets L, Prahl-Andersen B. A systematic review of the relationship between overjet size and traumatic dental injuries. Eur J Orthod 1999;21:503–15.	No assessment tool, original research
Rangel SJ, Kelsey J, Colby CE, Anderson J, Moss RL. Development of a quality assessment scale for retrospective clinical studies in pediatric surgery. J Pediatr Surg 2003;38:390–96.	Assessment tool tailored to a particular disease/topic
Reisch JS, Tyson JE, Mize SG. Aid to the evaluation of therapeutic studies. Pediatrics 1989;84:815–27.	No assessment tool
Stock SR. Workplace ergonomic factors and the development of musculoskeletal disorders of the neck and upper limbs: a metaanalysis. Am J Ind Med 1991;19:87–107.	No assessment tool, original research
van der Windt DAWM, Thomas E, Pope DP et al. Occupational risk factors for shoulder pain: a systematic review. Occup Environ Med 2000;57:433–42.	No assessment tool, original research
Margetts BM, Vorster HH, Venter CS. Evidence-based nutrition—review of nutritional epidemiological studies. South African J Clin Nutr 2002;15:68–73.	Assessment tool tailored to a particular disease/topic
Higgins JP, Altman DG, Gotzsche PC, Juni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011; 343: d5928.	Assessment of RCTs
Campos TF, Beckenkamp PR, Moseley AM. Usage evaluation of a resource to support evidence-based physiotherapy: the physiotherapy evidence database (PEDro). Physiotherapy 2013; 99(3): 252–57.	Assessment of RCTs

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Jadad AR, Moore RA, Carroll D, Jenkinson C, Reynolds DJ, Gavaghan DJ, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Controlled Clinical Trials 1996; 17(1): 1–12	Assessment of RCTs
Verhagen AP, de Vet HC, de Bie RA, Kessels AG, BoersM, Bouter LM, et al. The Delphi list: a criteria list for quality assessment of randomized clinical trials for conducting systematic reviews developed by Delphi consensus. Journal of Clinical Epidemiology 1998; 51(12): 1235–41.	Assessment of RCTs
Moga C, Guo B, Schopflocher D, Harstall C. Development of a Quality Appraisal Tool for Case Series Studies using a Modified Delphi Technique. 2012.	Assessment of case series studies
Stroke Therapy Academic Industry R. Recommendations for standards regarding preclinical neuroprotective and restorative drug development. Stroke 1999; 30(12): 2752–58.	Assessment of animal studies
Fisher M, Feuerstein G, Howells DW, Hurn PD, Kent TA, Savitz SI, et al. Update of the stroke therapy academic industry roundtable preclinical recommendations. Stroke 2009; 40(6): 2244–50.	Assessment of animal studies
Macleod MR, O'Collins T, Howells DW, Donnan GA. Pooling of animal experimental data reveals influence of study design and publication bias. Stroke 2004; 35(5): 1203–08.	Assessment of animal studies
Shea BJ, Grimshaw JM, Wells GA, Boers M, Andersson N, Hamel C, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. BMC Medical Research Methodology 2007; 7: 10.	Assessment of systematic reviews
Collaboration A. Development and validation of an international appraisal instrument for assessing the quality of clinical practice guidelines: the AGREE project. Quality & Safety in Health Care 2003; 12(1): 18–23	Assessment of clinical practice guidelines
Brouwers MC, Kho ME, Browman GP, Burgers JS, Cluzeau F, Feder G, et al. AGREE II: advancing guideline development, reporting and evaluation in health care. Canadian Medical Association Journal 2010; 182(18): E839–E842.	Assessment of clinical practice guidelines
Kirkham JJ, Dwan KM, Altman DG, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. BMJ 2010;340:c365.	Assessment of RCTs
Meader N, King K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. Systematic reviews 2014;3(1):82.	Assessment of systematic reviews
Dwan K, Gamble C, Kolamunnage-Dona R, et al. Assessing the potential for outcome reporting bias in a review: a tutorial. Trials 2010;11:52.	Assessment of RCTs
Stewart GB, Higgins JP, Schunemann H, et al. The use of Bayesian networks to assess the quality of evidence from research synthesis: 1. PLoS One 2015;10(3):e0114497.	Assessment of systematic reviews
Salanti G, Giovane CD, Chaimani A, et al. Evaluating the quality of evidence from a network meta-analysis. PLoS One 2014;9(7):e99682.	Assessment for network meta-analyses
Higgins JP, Del Giovane C, Chaimani A, et al. Evaluating the quality of evidence from a network meta-analysis. Value Health 2014;17(7):A324.	Assessment for network meta-analyses
WP5, EMA, EUnetHTA Joint Action 3 (2016 - 2020), REQueST® tool. Available at: https://eunethta.eu/wp-content/uploads/2019/10/EUnetHTAJA3_Vision_paper-on-REQueST-tool.pdf	Assessment specific on the quality of registries

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Figure S3. Assessment tools that included or not at least one question/item to address subdomains of methods for selecting study participants (Domain 1), stratified by study design.

	Domain 1 - Methods for selecting participants (SELECTION BIAS)				
Tools for Case-Control Studies only	Sampling strategies to correct selection bias	Inclusion and exclusion criteria of target population	Depletion of susceptibles	Generalizability of target population	Others^
RAMboMAN (GATE-EPIQ) 2019	×	✓	×	✓	✓
CASP 2018	✓	✓	×	✓	✓
SURE 2018	✓	✓	×	×	✓
JBI 2017	×	×	×	×	✓
NIH–NHLBI 2014	✓	✓	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	✓	✓
SIGN Version 3.0 2012	×	✓	×	✓	✓
STROBE statement V4 2007	✓	✓	×	×	✓
Durant 1994	×	×	×	✓	✓
Gyorkos 1994	×	×	×	×	✓
Lichtenstein 1987	✓	✓	×	×	✓
Horwitz 1979	✓	✓	×	×	✓
Tools for Cohort Studies only					
RAMboMAN (GATE-EPIQ) 2019	×	✓	×	✓	✓
CASP 2018	✓	×	×	✓	✓
SURE 2018	✓	✓	×	×	✓
JBI 2017	×	✓	×	×	✓
NIH–NHLBI 2014	×	✓	×	✓	✓
HEB Wales 2014	×	×	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	✓	✓
SIGN Version 3.0 2012	×	×	×	×	✓
STROBE statement V4 2007	✓	✓	×	×	✓
Hadorn 1996	✓	✓	×	✓	✓
Gyorkos 1994	×	×	×	×	✓
Tools for multiple NRS					
RELEVANT 2019	×	✓	×	✓	✓
MMAT 2018	×	✓	×	✓	×
ROBINS-I 2016	×	✓	×	×	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	×	×	✓
GRACE 2014 (#)	×	×	×	×	×
RoBANS 2013	×	×	×	×	✓
RTI 2013	✓	✓	×	×	✓
Montreal 2011	✓	×	×	✓	×
EPHPP 2004	×	×	×	✓	✓
TREND statement 2004	✓	✓	×	×	✓
Margetts 2002	✓	✓	×	×	✓
Zaza 2000	✓	✓	×	✓	✓
Downs-Black 1998	×	✓	×	✓	✓
Elwood 1998	×	✓	×	×	✓
London 1996	×	×	×	✓	✓
Avis 1994	✓	×	×	×	✓
Levine 1994	×	×	×	×	✓
Cho 1994 (#)	×	✓	×	×	✓
COEH 1991	✓	✓	×	✓	✓
Fowkes-Fulton 1991	✓	✓	×	✓	✓
Gardner 1986	×	×	×	×	✓

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
^ "Others" refers to items not included in our evaluation framework but included in the reviewed tools to investigate selection bias
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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Figure S4. Assessment tools that included or not at least one question/item to address subdomains for definition and measurement of exposure, outcomes, covariate variables and follow-up (Domain 2), stratified by study design.

	Domain 2 - Definition and measurement of exposure, outcomes, covariate variables and follow-up.			
Tools for Case-Control Studies only	Exposure	Outcome	Covariate variables	Follow-up
RAMboMAN (GATE-EPIQ) 2019	✓	✓	×	✓
CASP 2018	✓	✓	✓	×
SURE 2018	✓	✓	×	×
JBI 2017	✓	✓	×	✓
NIH–NHLBI 2014	✓	✓	×	×
Newcastle-Ottawa 2013	✓	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×	×
STROBE statement V4 2007	✓	✓	✓	✓
Durant 1994	×	×	✓	×
Gyorkos 1994	✓	✓	×	×
Lichtenstein 1987	✓	✓	×	×
Horwitz 1979	✓	✓	×	×
Tools for Cohort Studies only				
RAMboMAN (GATE-EPIQ) 2019	✓	✓	×	✓
CASP 2018	✓	✓	✓	✓
SURE 2018	✓	✓	×	✓
JBI 2017	✓	✓	×	✓
NIH–NHLBI 2014	✓	✓	×	✓
HEB Wales 2014	✓	✓	✓	✓
Newcastle-Ottawa 2013	✓	✓	✓	✓
SIGN Version 3.0 2012	✓	✓	×	×
STROBE statement V4 2007	✓	✓	✓	✓
Hadorn 1996	✓	✓	×	×
Gyorkos 1994	✓	✓	×	✓
Tools for multiple NRS				
RELEVANT 2019	✓	✓	×	×
MMAT 2018	✓	✓	×	×
ROBINS-I 2016	✓	✓	✓	×
ISPOR-AMCP-NPC 2014 (#)	✓	✓	×	✓
GRACE 2014 (#)	✓	✓	✓	×
RoBANS 2013	✓	✓	×	×
RTI 2013	✓	✓	✓	✓
Montreal 2011	×	✓	✓	×
EPHPP 2004	×	×	×	×
TREND statement 2004	✓	✓	×	✓
Margetts 2002	✓	✓	×	×
Zaza 2000	✓	✓	×	×
Downs-Black 1998	✓	✓	×	×
Elwood 1998	✓	✓	×	×
London 1996	✓	✓	×	×
Avis 1994	×	×	×	×
Levine 1994	✓	✓	×	✓
Cho 1994 (#)	×	×	×	×
COEH 1991	✓	×	×	×
Fowkes-Fulton 1991	✓	✓	×	×
Gardner 1986	×	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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Figure S5. Assessment tools that included or not at least one question/item to address subdomains to detect design-specific sources of biases (Domain 3), stratified by study design.

	Domain 3 - Design-specific sources of bias								
Tools for Case-Control Studies only	new user design	active comparator design	immortal time bias	detection or surveillance bias	loss to follow-up bias	Non-contemporaneous comparator bias	Reverse causation	Misclassification bias	Others^
RAMboMAN (GATE-EPIQ) 2019	×	×	×	×	×	×	×	✓	✓
CASP 2018	×	×	×	×	×	×	×	✓	✓
SURE 2018	×	×	×	×	×	×	×	×	×
JBI 2017	×	×	×	×	×	×	×	×	×
NIH–NHLBI 2014	×	×	×	×	×	✓	✓	×	×
Newcastle-Ottawa 2013	×	×	×	×	✓	×	×	×	×
SIGN Version 3.0 2012	×	×	×	×	×	×	×	×	✓
STROBE statement V4 2007	×	×	×	×	×	×	×	×	✓
Durant 1994	×	×	×	✓	×	×	×	×	✓
Gyorkos 1994	×	×	×	×	×	×	×	×	✓
Lichtenstein 1987	×	×	×	×	×	×	×	×	✓
Horwitz 1979	×	×	×	✓	×	×	×	×	✓
Tools for Cohort Studies only									
RAMboMAN (GATE-EPIQ) 2019	×	×	×	×	✓	×	✓	✓	×
CASP 2018	×	×	×	×	×	×	×	×	×
SURE 2018	×	×	×	×	✓	×	×	×	✓
JBI 2017	×	×	×	×	✓	×	✓	×	×
NIH–NHLBI 2014	×	×	×	×	✓	×	✓	×	×
HEB Wales 2014	×	×	×	×	✓	×	×	×	×
Newcastle-Ottawa 2013	×	×	×	×	✓	×	✓	×	×
SIGN Version 3.0 2012	×	×	×	✓	✓	×	✓	×	×
STROBE statement V4 2007	×	×	×	×	✓	×	×	×	×
Hadorn 1996	×	×	×	×	✓	×	×	×	×
Gyorkos 1994	×	×	×	×	×	×	×	×	×
Tools for multiple NRS									
RELEVANT 2019	×	×	×	×	×	×	×	×	×
MMAT 2018	×	×	×	×	✓	×	×	×	×
ROBINS-I 2016	✓	×	✓	×	×	×	×	×	×
ISPOR-AMCP-NPC 2014 (#)	×	×	×	×	×	✓	×	×	×
GRACE 2014 (#)	✓	×	✓	×	×	✓	×	×	×
RoBANS 2013	×	×	×	✓	✓	✓	×	×	✓
RTI 2013	×	×	×	×	✓	×	×	×	×
Montreal 2011	×	×	×	×	✓	×	×	×	×
EPHPP 2004	×	×	×	×	✓	×	×	✓	×
TREND statement 2004	×	×	×	×	×	×	×	×	×
Margetts 2002	×	×	×	×	×	×	×	×	×
Zaza 2000	×	×	×	×	✓	×	×	×	×
Downs-Black 1998	×	×	×	×	✓	✓	×	×	×
Elwood 1998	×	×	×	×	×	×	✓	×	×
London 1996	×	×	✓	×	✓	×	×	×	×
Avis 1994	×	×	×	×	✓	×	×	✓	×
Levine 1994	×	×	×	×	×	×	×	×	×
Cho 1994 (#)	×	×	×	×	✓	×	×	✓	×
COEH 1991	×	×	×	×	×	×	×	×	×
Fowkes-Fulton 1991	×	×	×	×	✓	×	×	×	×
Gardner 1986	×	×	×	×	×	×	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
^ "Others" refers to items on design-specific biases not included in the evaluation framework but addressed by the reviewed tools (e.g., recall bias, interviewer or observer bias, ascertment bias, etc.)
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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Figure S6. Assessment tools that included or not at least one question/item to address confounding (Domain 4), stratified by study design.

	Domain 4 - Confounding		
Tools for Case-Control Studies only	study design used to minimize the confounding	confounders measured and included in the analyses	potential unmeasured confounding addressed in the analyses
RAMboMAN (GATE-EPIQ) 2019	✓	×	×
CASP 2018	✓	✓	×
SURE 2018	×	✓	×
JBI 2017	✓	✓	×
NIH–NHLBI 2014	×	✓	×
Newcastle-Ottawa 2013	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×
STROBE statement V4 2007	×	✓	×
Durant 1994	✓	✓	✓
Gyorkos 1994	×	✓	×
Lichtenstein 1987	✓	✓	×
Horwitz 1979	×	×	×
Tools for Cohort Studies only			
RAMboMAN (GATE-EPIQ) 2019	✓	×	×
CASP 2018	✓	✓	×
SURE 2018	×	✓	×
JBI 2017	✓	✓	×
NIH–NHLBI 2014	×	✓	×
HEB Wales 2014	×	✓	✓
Newcastle-Ottawa 2013	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×
STROBE statement V4 2007	×	✓	×
Hadorn 1996	✓	✓	×
Gyorkos 1994	×	✓	×
Tools for multiple NRS			
RELEVANT 2019	✓	✓	×
MMAT 2018	✓	✓	×
ROBINS-I 2016	×	✓	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	✓
GRACE 2014 (#)	✓	✓	×
RoBANS 2013	✓	✓	×
RTI 2013	✓	✓	×
Montreal 2011	×	✓	×
EPHPP 2004	✓	✓	×
TREND statement 2004	✓	✓	×
Margetts 2002	×	×	×
Zaza 2000	✓	✓	✓
Downs-Black 1998	×	✓	×
Elwood 1998	✓	✓	×
London 1996	✓	✓	×
Avis 1994	×	✓	×
Levine 1994	×	×	×
Cho 1994 (#)	✓	✓	×
COEH 1991	×	✓	×
Fowkes-Fulton 1991	✓	✓	×
Gardner 1986	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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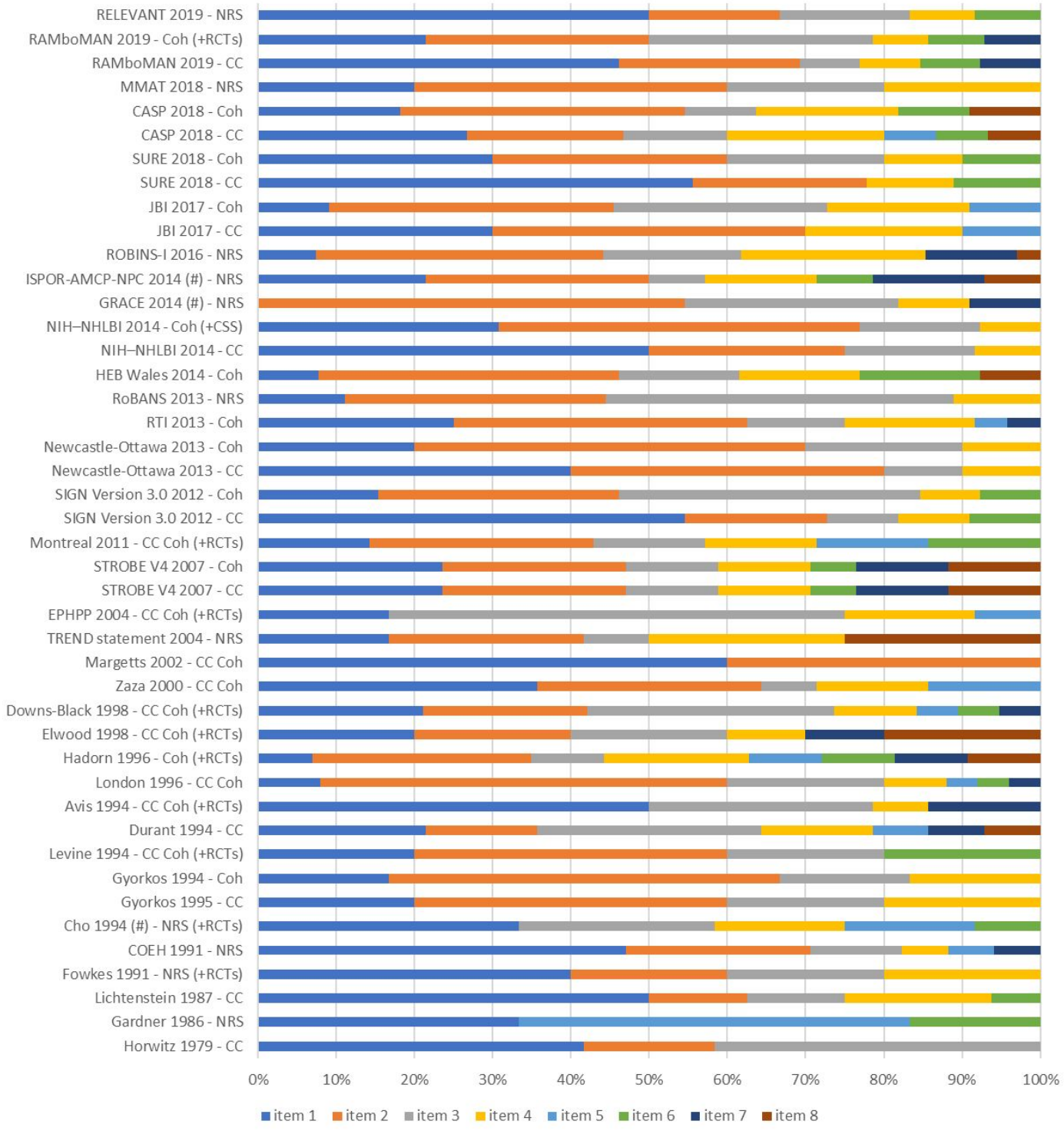
Figure S7. Assessment tools that included or not at least one question/item to address lack of appropriateness of statistical analyses (Domain 5), methods for assessing statistical uncertainty in the findings (Domain 6), methods for assessing internal validity (Domain 7), methods for assessing external validity (Domain 8), stratified by study design.

	Domain 5 - Lack of appropriate statistical analyses	**Domain 6 - Statistical uncertainty**	**Domain 7 - Internal validity**	**Domain 8 - External validity**
Tools for Case-Control Studies only	(e.g., overadjustment for intermidiates, incorrect outcome model specification)	(e.g., confidence intervals reported for each analysis)	(e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases)	(e.g. post-hoc subgroup analysis, comparison with other populations etc.)
RAMboMAN (GATE-EPIQ) 2019	×	✓	✓	×
CASP 2018	✓	✓	×	✓
SURE 2018	×	✓	×	×
JBI 2017	✓	×	×	×
NIH–NHLBI 2014	×	×	×	×
Newcastle-Ottawa 2013	×	×	×	×
SIGN Version 3.0 2012	×	✓	×	×
STROBE statement V4 2007	×	✓	✓	✓
Durant 1994	✓	×	✓	✓
Gyorkos 1994	×	×	×	×
Lichtenstein 1987	×	✓	×	×
Horwitz 1979	×	×	×	×
Tools for Cohort Studies only				
RAMboMAN (GATE-EPIQ) 2019	×	✓	✓	×
CASP 2018	×	✓	×	✓
SURE 2018	×	✓	×	×
JBI 2017	✓	×	×	×
NIH–NHLBI 2014	×	×	×	×
HEB Wales 2014	×	✓	×	✓
Newcastle-Ottawa 2013	×	×	×	×
SIGN Version 3.0 2012	×	✓	×	×
STROBE statement V4 2007	×	✓	✓	✓
Hadorn 1996	✓	✓	✓	✓
Gyorkos 1994	×	×	×	×
Tools for multiple NRS				
RELEVANT 2019	×	✓	×	×
MMAT 2018	×	×	×	×
ROBINS-I 2016	×	×	✓	✓
ISPOR-AMCP-NPC 2014 (#)	×	✓	✓	✓
GRACE 2014 (#)	×	×	✓	×
RoBANS 2013	×	×	×	×
RTI 2013	✓	×	✓	×
Montreal 2011	✓	✓	×	×
EPHPP 2004	✓	×	×	×
TREND statement 2004	×	×	×	✓
Margetts 2002	×	×	×	×
Zaza 2000	✓	×	×	×
Downs-Black 1998	✓	✓	✓	×
Elwood 1998	×	×	✓	✓
London 1996	✓	✓	✓	×
Avis 1994	×	×	✓	×
Levine 1994	×	✓	×	×
Cho 1994 (#)	✓	✓	×	×
COEH 1991	✓	×	✓	×
Fowkes-Fulton 1991	×	×	×	×
Gardner 1986	✓	✓	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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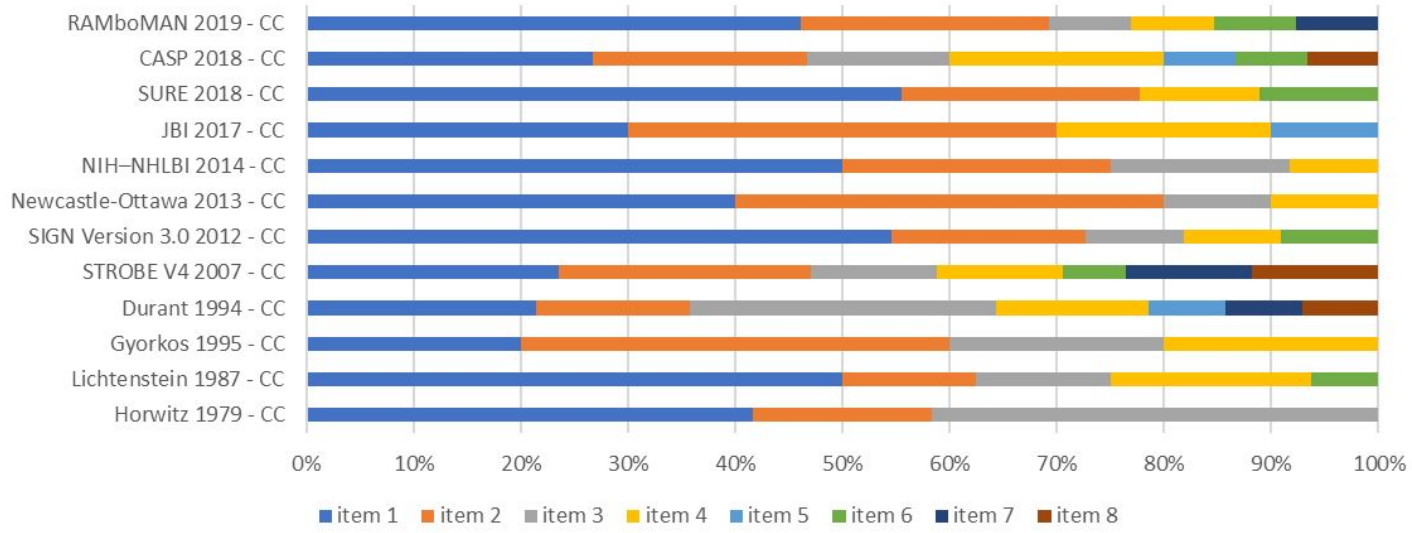
Figure S8. Distribution of domains for each reviewed tool



Legend: Item 1 = Methods for selecting participants; item 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; item 3 = Design-specific sources of bias; item 4 = Confounding; item 5 = Lack of appropriateness of statistical analyses; item 6 = Methods for assessing statistical uncertainty in the findings; item 7 = Methods for assessing internal validity; item 8 = Methods for assessing external validity.

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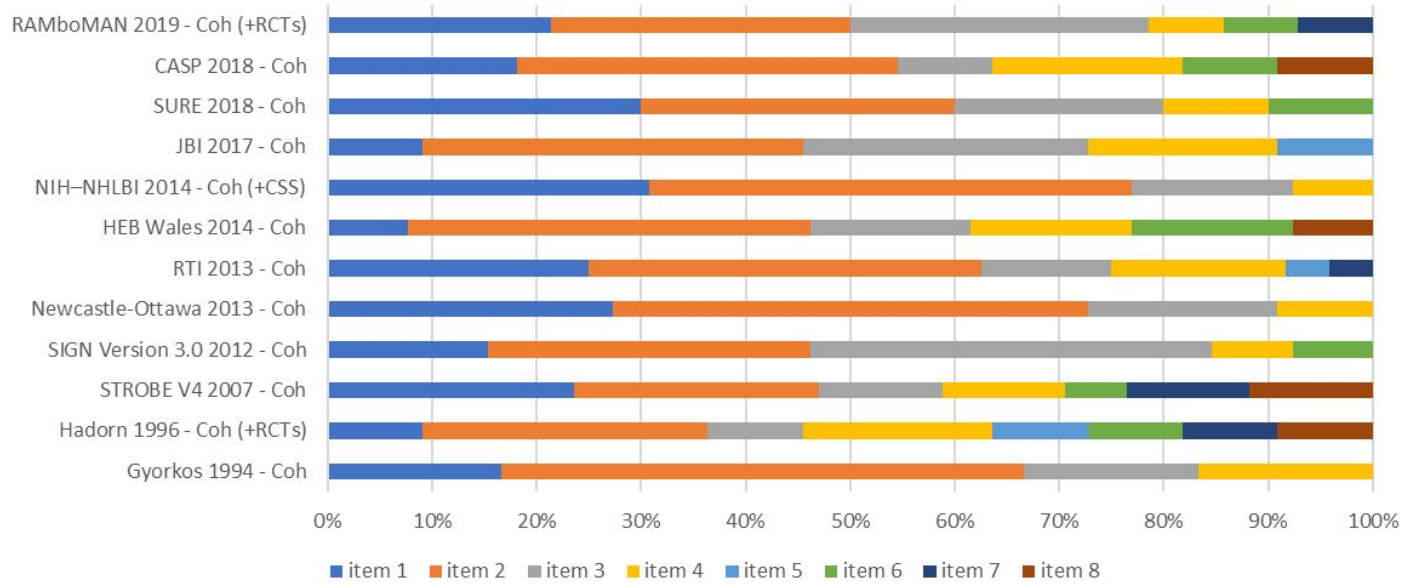
Figure S9. Distribution of domains for each reviewed tool of case-control studies



Legend: Item 1 = Methods for selecting participants; item 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; item 3 = Design-specific sources of bias; item 4 = Confounding; item 5 = Lack of appropriateness of statistical analyses; item 6 = Methods for assessing statistical uncertainty in the findings; item 7 = Methods for assessing internal validity; item 8 = Methods for assessing external validity.

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Figure S10. Distribution of domains for each reviewed tool of cohort studies

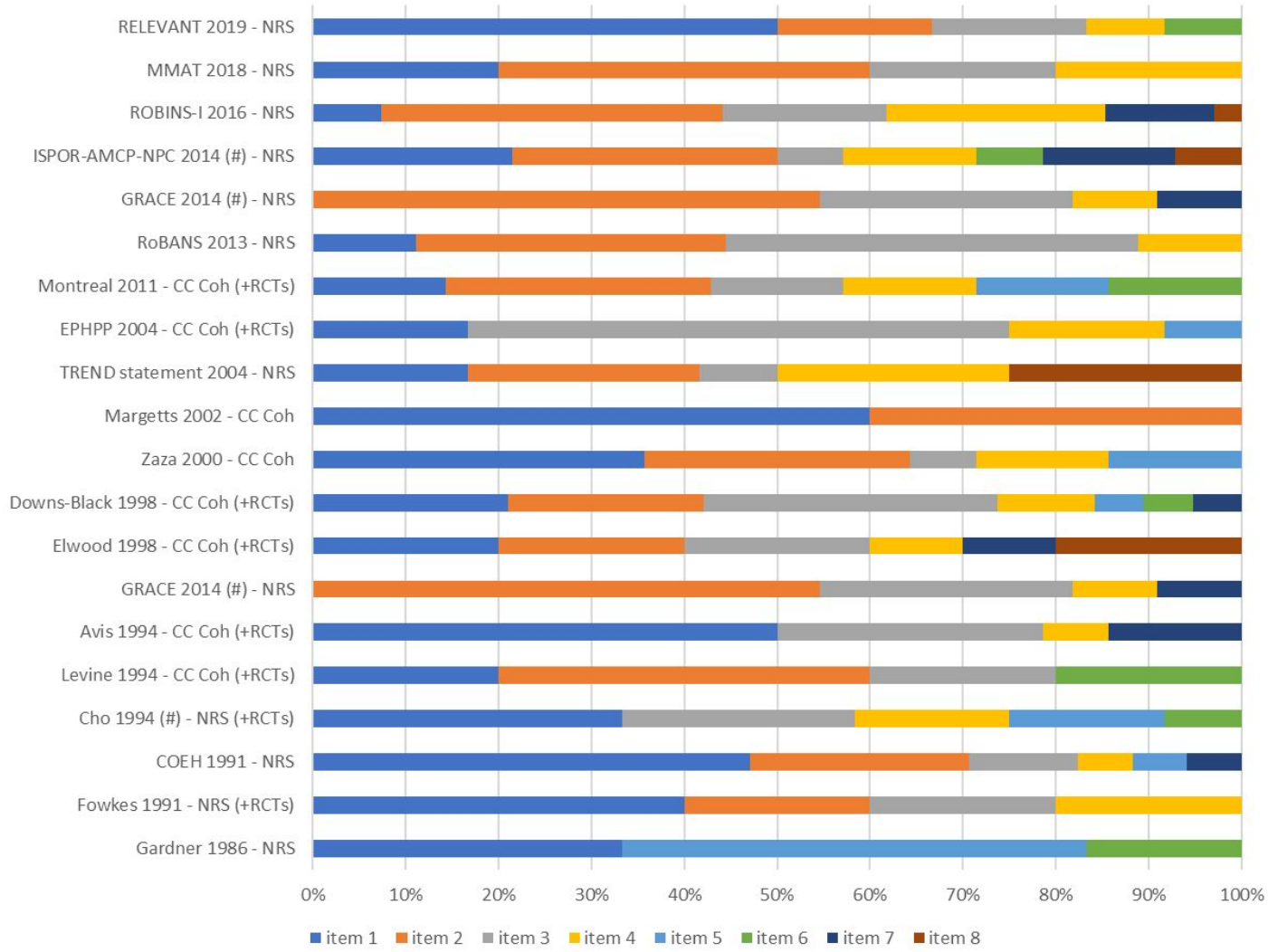


Legend: Item 1 = Methods for selecting participants; item 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; item 3 = Design-specific sources of bias; item 4 = Confounding; item 5 = Lack of appropriateness of statistical analyses; item 6 = Methods for assessing statistical uncertainty in the findings; item 7 = Methods for assessing internal validity; item 8 = Methods for assessing external validity.

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Figure S11. Distribution of domains for each reviewed tool of multiple study designs (NRS)



Legend: Item 1 = Methods for selecting participants; item 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; item 3 = Design-specific sources of bias; item 4 = Confounding; item 5 = Lack of appropriateness of statistical analyses; item 6 = Methods for assessing statistical uncertainty in the findings; item 7 = Methods for assessing internal validity; item 8 = Methods for assessing external validity.

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Reference

Hasson F, Keeney S, McKenna H. Research guidelines for the Delphi survey technique. J Adv Nurs. 2000;32(4):1008-1015.

Sørensen HT, Lash TL, Rothman KJ. Beyond randomized controlled trials: a critical comparison of trials with nonrandomized studies. Hepatology. 2006;44(5):1075-1082

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Supplement file 2

Survey to the experts of the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research (CER) - Special Interest Group: FRAMEWORK TO EVALUATE ASSESSMENT TOOLS

Dear Member,

As an expert contributor to the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research (CER) Special Interest Group, we would like to invite you to take part in a short survey aimed at identifying key quality domains that:
(i) should be considered when assessing the overall quality of a nonrandomized study (NRS) in pharmacoepidemiology, and;
(ii) are not covered by existing appraisal tools for assessing the quality of NRSs.

Your answers will be used to inform the research project on an evaluation of existing tools to assess the methodological quality of pharmacoepidemiological NRSs - specifically cohort and case-control studies.

The survey will take approximately 5 minutes to complete and consists of two sections:
A. Demographic and professional information
B. Evaluation of key quality domains

In SECTION A, you will be asked to provide your demographic and professional information.

SECTION B lists key domains in assessing the overall quality of NRSs in pharmacoepidemiology. Each key domain is presented as an item that can be part of a tool for evaluating the methodological quality of pharmacoepidemiological NRSs (cohort and case control designs). You are asked to evaluate the importance of each item, rating it from 1 (not important) to 5 (extremely important). At the end of section B, you will have the opportunity to report additional items that were not presented in the survey, are key for evaluating the methodological quality of pharmacoepidemiological NRSs and are not included in current appraisal tools.

Your answers will be kept strictly confidential and will be used only for the purposes of this research project. The results will be reported in aggregate form only and will be part of a publication in a peer-reviewed journal.

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Should you have any questions on the survey or the project, please contact Elvira D'Andrea at edandrea@bwh.harvard.edu or Lydia Vinals at lydia.vinals@evidera.com.

We thank you for your collaboration.

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Thomas Debray (t.debray@umcutrecht.nl)

Research Project coordinators.

Elisabetta Patorno (epatorno@bwh.harvard.edu)

Chair, ISPE CER Special Interest Group.

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Section A – Demographic and professional info

Years of experience in Pharmacoepidemiology and/or Healthcare Outcome Research _____

Institution or Organization:

– Academia

– Industry

– Governmental Agencies

– Other (specify) _____

Professional position _____

City _____

Country _____

Section B - Evaluation of key quality domains

Please rate the following items from 1 (not important) to 5 (extremely important). Your judgment should be based on the importance that the key domains described by each item should have in tools assessing the overall methodological quality of a pharmacoepidemiologic NRS.

How important is it for a tool to assess:

1. Methods for selecting participants (selection bias)

Not important	1	2	3	4	5	Extremely important

Specifically:

1.1 Sampling strategies to correct selection bias

Not important	1	2	3	4	5	Extremely important

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2.1 Inclusion and exclusion criteria of target population clearly defined

Not important	1	2	3	4	5	Extremely important

2.2 Depletion of susceptibles

Not important	1	2	3	4	5	Extremely important

2.3 Other key source of selection bias not included above _____

2. Definition and measurement of exposure, outcomes, covariate variables and follow-up.

Not important	1	2	3	4	5	Extremely important

Specifically:

2.1 Definition and measurement of exposure

Not important	1	2	3	4	5	Extremely important

2.2 Definition and measurement of outcomes

Not important	1	2	3	4	5	Extremely important

2.3 Definition and measurement of covariate variables

Not important	1	2	3	4	5	Extremely important

2.4 Definition and measurement of follow-up

Not important	1	2	3	4	5	Extremely important

3. Design-specific sources of bias:

Not important	1	2	3	4	5	Extremely important

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Specifically:

3.1 lack of use of new user design:

Not important	1	2	3	4	5	Extremely important

3.2 lack of use of active comparator:

Not important	1	2	3	4	5	Extremely important

3.3 Time-related biases (immortal time bias or window-time bias):

Not important	1	2	3	4	5	Extremely important

3.4 Detection or surveillance bias:

Not important	1	2	3	4	5	Extremely important

3.5 Loss to follow-up bias:

Not important	1	2	3	4	5	Extremely important

3.6 Non-contemporaneous comparator bias:

Not important	1	2	3	4	5	Extremely important

3.7 Reverse causation:

Not important	1	2	3	4	5	Extremely important

3.8 Recall bias:

Not important	1	2	3	4	5	Extremely important

3.9 Other critical design-specific sources of bias not mentioned above _____

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4. Confounding:

Not important	1	2	3	4	5	Extremely important

Specifically:

4.1 Study design used to minimize the confounding:

Not important	1	2	3	4	5	Extremely important

4.2 Confounders measured and included in the analyses

Not important	1	2	3	4	5	Extremely important

4.3 Potential unmeasured confounding (addressed in the analysis through proxy analysis, IV analysis, etc.)

Not important	1	2	3	4	5	Extremely important

4.4 Other critical elements for confounding not mentioned above _____

5. Lack of appropriateness of statistical analyses (with specific mention of adjustment for causal intermediaries, and/or incorrect outcome model specification)

Not important	1	2	3	4	5	Extremely important

6. Methods for assessing statistical uncertainty in the findings (e.g., confidence intervals reported for each analysis)

Not important	1	2	3	4	5	Extremely important

7. Methods for assessing internal validity (e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases)

Not important	1	2	3	4	5	Extremely important

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8. Methods for assessing external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.)

Not important	1	2	3	4	5	Extremely important

9. Any other key considerations not listed above _____

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GLOSSARY

SELECTION BIAS defined as a bias due to a sample selection that does not accurately reflect the target population.

DEPLETION OF SUSCEPTIBLES is defined as a phenomenon where the initiation of exposure to a drug is associated with an early increase in the incidence rate of an acute event, followed by a decrease in this incidence rate with longer duration of exposure.

NEW USER DESIGN, defined as a study design that starts following patients at the time they initiate a new drug (also known as incident user design).

ACTIVE COMPARATOR DESIGN, defined as a study design that compares the effect of the drug of interest with another drug used in clinical practice instead of non-use.

IMMORTAL TIME BIAS, defined as a bias that derives from including a period of follow-up during which, by design, outcomes cannot occur.

WINDOW-TIME BIAS, in the context of a case-control study nested in a cohort, defined as a bias that derives from the use of time-windows of different lengths between cases and controls to define time-dependent exposures.

DETECTION OR SURVEILLANCE BIAS, defined as a bias that derives from systematic differences in outcomes assessment among groups being compared.

NON-CONTEMPORANEOUS COMPARATOR BIAS, defined as a bias generated by differences in the timing of selection of comparator group(s) within a study influence exposures and outcomes resulting in biased estimates.

REVERSE CAUSATION (or reverse causality), defined as a direction of cause-and-effect contrary to a common presumption, or a two-way causal relationship between exposure and outcome.

RECALL BIAS, defined as a bias that occurs when participants do not remember previous events or experiences accurately or omit details (not for claims-based studies).

LOSS TO FOLLOW-UP BIAS, defined as a bias that occurs when there is difference in retention during the follow-up period after enrollment that are related to exposure status and outcome. It can be considered as part of selection bias.

CONFOUNDING, defined as a mixing of effects that arises when patients with different baseline risks are compared – the resulting effect measure is a mix of drug effects and risk factor effects.

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ADJUSTING FOR CAUSAL INTERMEDIARIES, defined as adjustment for an intermediate variable (or a descending proxy for an intermediate variable) on a causal path from exposure to outcome.

INCORRECT OUTCOME MODEL SPECIFICATION, for omission of a relevant variable, inclusion of an unnecessary variable, adopting the wrong functional form, incorrect specification of the error term, uncertainty about what the true model is, reciprocal causation, etc.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	# 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	# 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	# 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	# 4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	# 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	# 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	# 5 and Suppl. 1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	# 5 and Suppl. 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	# 6-7 and Suppl. 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	# 5-6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	# 5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	# 5-6 (framework)
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	# 6

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PRISMA 2009 Checklist

Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	---

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	---
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	# 6 and Suppl. 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	# 7-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	# 7-9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Suppl. 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	---
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Suppl. 1
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	# 7-9
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	# 9-10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	# 12-13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	# 13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	# 13-14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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How well can we assess the validity of non-randomized studies of medications? A systematic review of assessment tools by the Working Group of the ISPE Comparative Effectiveness Research Special Interest Group

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Title: How well can we assess the validity of non-randomized studies of medications? A systematic review of assessment tools by the Working Group of the ISPE Comparative Effectiveness Research Special Interest Group

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Abstract

Objective: To determine whether assessment tools for non-randomized studies (NRS) address critical elements that influence the validity of NRS findings for comparative safety and effectiveness of medications.

Design: Systematic review and Delphi survey.

Data Sources: We searched Pubmed, Embase, Google®, bibliographies of reviews and websites of influential organizations from inception to November 2019. In parallel, we conducted a Delphi survey among the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research Special Interest Group to identify key methodological challenges for NRS of medications. We created a framework consisting of the reported methodological challenges to evaluate the selected NRS tools.

Study selection: Checklists or scales assessing NRS.

Data extraction: Two reviewers extracted general information and content data related to the prespecified framework.

Results: Of 44 tools reviewed, 48% (n = 21) assess multiple NRS designs, while other tools specifically addressed case-control (n = 12, 27%) or cohort studies (n = 11, 25%) only. Response rate to the Delphi survey was 73% (35 out of 48 content experts) and a consensus was reached in only two rounds. Most tools evaluated methods for selecting study participants (n = 43, 98%), although only one addressed selection bias due to depletion of susceptibles (2%). Many tools addressed the measurement of exposure and outcome (n = 40, 91%), and measurement and control for confounders (n = 40, 91%). Most tools have at least one item/question on design-specific sources of bias (n = 40, 91%), but only a few investigate reverse causation (n = 8, 18%), detection bias (n = 4, 9%), time-related bias (n = 3, 7%), lack of new-user design (n = 2, 5%), or active comparator design (n = 0). Few tools address the appropriateness of statistical analyses (n = 15, 34%), methods for assessing internal (n = 15, 34%) or external validity (n = 11, 25%) and statistical uncertainty in the findings (n = 21, 48%). None of the reviewed tools investigated all the methodological domains and subdomains.

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Conclusions: The acknowledgement of major design-specific sources of bias (e.g., lack of new-user design, lack of active comparator design, time-related bias, depletion of susceptibles, reverse causation), and statistical assessment of internal and external validity is currently not sufficiently addressed in most of the existing tools. These critical elements should be integrated to systematically investigate the validity of NRS on comparative safety and effectiveness of medications.

Systematic review protocol and registration https://osf.io/es65q

Keywords: systematic review, real world data, real world evidence, tools, non-randomized studies, cohort, case-control, comparative effectiveness research, validity.

Strengths and limitations of this study

- This is the first systematic review to investigate whether existing tools adequately assess the validity of nonrandomized studies (NRS) evaluating the comparative safety and effectiveness of medications.
- Assessment tools were identified by searching through multiple sources: relevant databases, grey literature, websites of authoritative organizations, bibliographies of previous systematic reviews and experts' suggestions.
- The pre-piloted framework adopted to evaluate the completeness of the tools included all the main methodological challenges suggested by an interdisciplinary (academia, industry and government agencies) and international team of experts in the field of pharmacoepidemiology and healthcare outcomes research.
- Tools not published in English or that could not be retrieved were omitted from this systematic review.
- The search for tools in the grey literature might not be comprehensive since it was performed through only one browser.

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Introduction

There are high expectations that real-world data (RWD) and resultant real-world evidence (RWE) will become a key source of information for the development process of pharmacological or biological therapies [1-3]. The 21st Century Cures Act and the sixth Prescription Drug User Fee Act (PDUFA VI) required the Food and Drug Administration (FDA) to explore the use of RWE and, consequently, well-designed and conducted non-randomized studies (NRS) for expediting drug approvals [4,5]. Similarly, one of the goals of the European Medicines Agency (EMA) Adaptive Pathways Initiative is to supplement clinical trial data with RWD and to eventually produce RWE as part of the approval process of new medications or indications [6].

However, the growing demand for RWD has raised concerns about the reliability of NRS to generate RWE. Due to the inherent limitations of observational analyses, the validity of NRS depends largely on the implementation of complex design and analytic methodologies. In recent reports, both FDA and EMA emphasized the need to plan and execute NRS following standards that can ensure validity and reproducibility of RWE [7,8]. Tools that assess the validity of NRS can be useful instruments for both researchers (e.g., for authors and reviewers to prevent publication of poor quality pharmacoepidemiologic research) and other stakeholders who are involved in clinical, managemental, or economic decision-making (e.g., to correctly inform guidelines and clinicians, or to guide resource allocation).

An analysis on the capability of existing tools to assess the validity of NRS of comparative safety and effectiveness of medications is currently lacking. Previously published systematic reviews on assessment tools for NRS were mostly descriptive and did not provide a critical evaluation of the tools content [9-13], investigated only a specific type of bias [14] or focused only on safety outcomes [15]. Therefore, we conducted a systematic review to assess the content of eligible tools for NRS of medications. There is no agreement on an assessment framework for NRS of pharmacological interventions. Thus, we performed a Delphi survey among international experts in the field of *pharmacoepidemiology and health outcome research* in order to build consensus for the methodological

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challenges that may threaten the validity of NRS of medications and that should be evaluated by assessment NRS tools.

The main objective of this study was to determine whether the retrieved NRS tools sufficiently address the main methodological challenges recommended by the experts. This study is part of a research project to develop a framework for the synthesis of NRS and randomized controlled trials (RCT) [16], led by the Comparative Effectiveness Research Special Interest Group (CER SIG) of the International Society for Pharmacoepidemiology (ISPE).

Methods

The systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statement [17]. Systematic review protocol and registration are available at https://osf.io/es65q.

Systematic search and eligibility criteria

We searched Pubmed and Embase from inception to November 2019 to identify existing tools that investigated the validity of NRS, specifically case-control and cohort design studies. We excluded guidelines or manuals, tools to review study protocols, tools targeting NRS of non-pharmacological interventions (e.g. surgery) or assessing only one or a few specific types of bias, and tools not available in English language. In parallel, we searched the same electronic databases for systematic reviews of assessment tools of NRS. We then extracted the references of the tools included in the systematic reviews retrieved. We also performed a general search through Google® for grey literature and reviewed any additional information from initiatives, programs or organizations. Full details on the search strategy are reported in the Supplement (Table S1 and S2, online Supplement 1). Three reviewers (E.D., G.S., L.V.) independently removed duplicates and reviewed titles and abstracts of peer-reviewed publications or documents from the grey literature to select eligible tools. Discrepancies were resolved by consensus.

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Delphi survey and prespecified framework

Concurrently, we performed a Delphi survey [18] to reach a consensus among content experts about the main methodological challenges (domains) that may threaten the validity of NRS on comparative safety and effectiveness of medications. The survey is available in the online Supplement 2. The panel of experts involved members of the SIG for CER of the ISPE. Detailed information on the Delphi methods and results is reported in the online Supplement 1.

Domains and subdomains indicated by the Delphi respondents as major elements that can impact the validity of NRS of medications were used to develop and pilot a framework to evaluate the identified NRS tools. All domains were considered equally important. A glossary of terms used in the framework is reported in Table 1.

Data extraction

Two reviewers (E.D., L.V.) independently extracted general information of the identified tools (first author or name of the tool, year of publication or online availability of the most updated version, type of tool, scope of the tool, non-randomized study designs evaluated, number of items) and content data related to the prespecified domains of the framework. Discrepancies were resolved by consensus. We categorized the tools as checklists, defined as itemized instruments (including questionnaires) developed to identify the presence or absence of critical elements, or rating scales, defined as itemized instruments aimed to identify the performance of a study at each critical element described in the tool, using a qualitative or quantitative scale.

Data synthesis

General characteristics of the identified tools were summarized with means and standard deviations, for continuous variables, and relative frequencies, for categorical variables. The findings from

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the Delphi survey and the proportion of tools assessing the prespecified elements of the framework were reported in terms of relative frequencies.

Results

Overview of Tools

Of 44 tools that met our eligibility criteria [19-53], 20 (45%) were identified through the database search of peer-reviewed literature and 24 (55%) through the general online search and other sources (Figure S1 and Table S3 in the online Supplement 1). Characteristics of the tools are shown in Table 2 and Table 3. The number of items across all tools ranged from 5 to 54, with a median of 13.5 (IQR, 10.3-22). Only three tools were designed to specifically address studies on the comparative safety and effectiveness of pharmacological interventions: one published in 1994 by Cho et al. [46], the GRACE checklist and the ISPOR-AMCP-NPC tool, both published in 2014 [25,26].

Tool Formats and Scopes

Most of the tools were checklists (n = 35, 80%), and thirteen checklists included a final section to elaborate a summary judgement of the study appraisal (37%). The remaining tools were scales (n = 9, 20%), and six of them provided a section for a summary judgement (67%).

Thirty-five tools (80%) were designed as critical appraisal tools for different scopes (e.g., assessing the quality of NRS included in a systematic review, screening eligible NRS to include in systematic reviews to support clinical guidelines, supporting peer-review processes or, more general, allowing readers to interpret NRS results critically). Four tools (9%) were developed to assess the quality of reporting and were mainly intended for researchers. Five other tools (11%) combined elements of both critical appraisal and quality reporting and were for a more general audience (both researchers and readers) (Table 2 and Table 3).

Study Designs Addressed

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Twenty-one tools (48%) were developed to assess multiple NRS designs (11 targeted cohort and case-control studies and ten others addressed also other NRS designs or did not specify them). Other tools specifically addressed case-control (n = 12, 27%) or cohort studies (n = 11, 25%). Ten tools (23%) were designed to assess also RCTs.

Tool Elements

The response rate of the Delphi survey was 73% (35 respondents out of 48 members). Detailed results are reported in the Supplement 1 (Figure S2). Domains and subdomains indicated by the respondents as major elements that can impact the validity of NRS of medications are reported in the first column of Table 4.

Methods for selecting participants

Nearly all tools assessed methods for selecting study participants to correct selection bias (n = 43, 98%). Specifically, almost half of the tools included items related to sampling strategies (n = 19, 43%), the definition of inclusion and exclusion criteria (n = 27, 61%), and the generalizability of participants (i.e., attempts to achieve a sample of participants that represents the target population) (n = 21, 48%), while only one tool addressed the depletion of susceptibles (n = 1, 2%) (Table 4 and Figure S3 in the online Supplement).

Measurement of exposure, outcomes, covariates, and follow-up

Forty-two tools (95%) had at least one item assessing the definition and measurement of exposure, outcome, covariates, and follow-up. Assessment of exposure and outcome was widely reported by the tools (n = 40, 91%), while definition and measurement of covariates (n = 12, 27%) or follow-up (n = 17, 39%) were less often addressed (with the exception for tools addressing follow-up in cohort studies only, n = 9, 82%) (Table 4 and Figure S4 in the online Supplement).

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Design-specific sources of bias

Design-specific sources of bias (excluding selection bias which was investigated in "*Methods for selecting participants"*) were assessed by 91% of the tools (n = 40), and generally included loss to follow-up bias (n = 22, 50%), observer or interviewer bias (n = 11, 25%), reverse causation bias (n = 8, 18%), recall bias (n = 6, 14%), and non-contemporaneous comparator bias (n = 6, 14%). A few or no tools assessed detection or surveillance bias (n = 4, 9%), time related bias, such as immortal person-time bias or time-window bias (n = 3, 7%), and biases due to lack of new-user design (n = 2, 5%) or active comparator design (n = 0). Other tools reported only a general item/question on the risk of bias (n = 9, 20%), without any reference to a specific type of bias.

Tools specifically for cohort studies addressed more frequently loss to follow-up (n = 9, 82%) and reverse causation biases (n = 5, 45%) compared to the other tools, while tools for case-control studies addressed mostly recall (n = 4, 33%) and observer biases (n = 3, 25%). Tools for multiple NRS covered commonly loss to follow up (n = 12, 57%) and interviewer or observer biases (n = 7, 35%) (Table 4 and Figure S5 in the online Supplement).

Confounding

Forty tools (91%) included at least one item or question related to confounding. Specifically, 26 tools (59%) searched whether study design was planned in a way to minimize confounding, 38 (86%) whether confounders were measured and included in the analyses, and only five whether potential unmeasured confounding was assessed in the sensitivity analyses (11%) (Table 4 and Figure S6 in the online Supplement).

Appropriateness of statistical analyses, external and internal validity

One-third of the tools (n = 14, 32%) assessed the appropriateness of statistical analyses, although most of them did not explicitly mention overadjustment of causal intermediates and/or incorrect outcome model specification. Almost half (n = 21, 48%) included methods for measuring uncertainty in the

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findings. Few tools addressed methods for evaluating internal (n = 15, 34%) or external validity (n = 11, 25%) (Table 4 and Figure S7 in the online Supplement).

These results were mostly consistent across the three different types of design addressed, cohort only, case-control only and multiple NRS, except for the assessment of follow-up (Domain 2) and several design-specific sources of biases (Domain 3) already mentioned above (Table 4). None of the reviewed tools covered all the main domains and subdomains as identified by the CER SIG and listed in Table 4.

Results for each selected tool on the proportions of items/questions that investigate the prespecified domains are shown in the Figures S8-S11 in the Supplement 1.

Discussion

In this systematic review, we identified assessment tools evaluating the validity of NRS on comparative safety and effectiveness of medications. Of 44 tools included, only three were specifically designed to assess NRS of pharmacological interventions [25,26,46].

Main findings

Overall, we found that existing tools assessed most of the methodological challenges identified by the domains of the CER SIG framework, but critical elements were often insufficiently addressed. For example, although many tools assessed the risk of selection bias, only half of them explicitly investigated sampling strategies and considered a pre-specification of inclusion/exclusion criteria. Even more surprising was that only one tool explored the potential for selection bias due to depletion of patients that are susceptible to the outcome. This cohort-based phenomenon can occur when new users of a medication are depleted of all susceptible subjects to the outcome, documenting an increased incidence rate of the outcome in an early stage, followed by a decreased rate with a longer duration of exposure [54]. Depletion of susceptibles is an important source of bias to account for when evaluating effects of new medications in incident users and can significantly undermine the validity of the results [54].

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Similarly, many tools investigated misclassification or information bias of exposure and outcome. However, only about one-third assessed definition and measurement of covariates, and less than one-fourth of the case-control and multiple-NRS designs tools assessed information on follow-up definition. Again, these are common causes of bias and should be integrated in tools that investigate the validity of NRS.

Design-specific sources of bias was a critical domain. Although overall 91% of the tools had at least one item/question investigating biases due to an inappropriate study design, only ROBINS-I and the GRACE checklist addressed bias due to lack of new user design, and time-related bias (i.e., immortal person-time bias or time-window bias), while no tools investigated bias due to lack of active comparator design. Since these biases can independently lead to major methodological flaws (defined as elements that by themselves can significantly compromise the validity of the results), their assessment must be included in appraisal tools for NRS of pharmacological interventions. For example, recent evidence on NRS of glucose-lowering medications reported that only one-fourth of the studies adopted a new-user design and less than half used an active comparator [55]. In the same example, potential for time-related bias was detected in more than two-third of the studies [55]. Integrating the evaluation of these major methodological flaws in existing tools and recommending the use of these tools before publication can increase awareness in the clinical research community with respect to main design-specific biases. This can ultimately decrease the amount of NRS with invalid findings on the safety or effectiveness of medications.

A high percentage of tools evaluated whether confounders were appropriately measured, controlled for in the analysis, and considered in the study design. However, very few tools included at least one item/question on whether potential unmeasured confounding had been considered in the analysis or interpretation of findings.

One-third of the tools checked the appropriateness of statistical analyses, but most omitted specific reference to common flaws such as overadjustment or incorrect outcome model specification. Similarly, only one-third of the tools assessed internal validity (e.g., through sensitivity analysis to

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address potential confounding, measurement errors or other biases), and only one-fourth assessed external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.).

Implications for practice and research

While recently published tools such as the CASP checklist [21], ISPOR-AMCP-NPC [25], RAMboMAN [19], GRACE [26] and ROBINS-I [24] are among the most complete tools, addressing several of the critical elements underlined by the ISPE CER SIG, they all had limitations in the acknowledgement of two or more major methodological challenges (e.g., selection bias for depletion of susceptibles, immortal-time bias or window-time bias, lack of new user design, lack of active comparator design, reverse causation bias, adjustment for causal intermediaries). Assessment tools can be powerful instruments for researchers, authors, reviewers of scientific journals or readers, helping to identify the main limitations of a study and to correctly interpret the results, to acknowledge major methodological flaws, and, ultimately, to prevent publication of studies with invalid findings.

Further, other decision-makers, such as clinicians, guideline developers, and payers or investors, can benefit from instruments that help to ensure the validity of NRS findings. RCTs can be an insufficient source of evidence for decisions on pharmaceutical interventions [56, 57]. Despite well-designed and adequately powered RCTs being considered the “gold standard” of the clinical research paradigm, they can often be too time- and money-intensive. Trials are often relatively small, focus on short-term efficacy and safety in a controlled clinical environment, using surrogate outcomes, or underrepresenting high-risk populations that can be most likely the target on the new medications in the real-world setting [56, 57]. Trials might also not record treatments taken outside the study protocol [49]. Additionally, patients volunteered to participate in a trial are usually very motivated and so more adherent to therapy compared to the real-world population [57]. NRS based on RWD can help to address these issues and could be supplement the evidence from RCTs to provide a more complete picture on the effectiveness of pharmaceutical interventions in less controlled environments. NRS have the advantages to investigate large-scale populations, high-risk subpopulations, rare exposures, diseases or outcome, and long-term

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outcomes or other delayed health effects at low costs and rapidly [56, 57]. Moreover, since RWD are often collected for intents unrelated to research objectives (mainly administrative), biases such as recall bias, interviewer bias, nonresponse bias and bias for loss to follow-up are reduced or eliminated [56]. Thus, since RWE derived by NRS contribute significantly to generate evidence of comparative effectiveness research of medications, our synthesis can help numerous stakeholders to evaluate whether the NRS considered are valid enough to guide decision-making.

Although checklists have been previously suggested for reviewing the risk of bias of general NRS [58], we cannot strongly recommend a specific tool for NRS on comparative analyses of medications. As already mentioned, items or questions that address all those methodological flaws must be integrated in the existing tools. Based on our findings, most recent and comprehensive tools such as ROBINS-I [24] and GRACE [26] assessed a higher number of major methodological elements and could therefore be prioritized in this endeavor.

Strengths and limitations

To our knowledge this is the first systematic review that investigated whether existing tools adequately assess the validity of cohort and case-control studies evaluating the comparative safety and effectiveness of medications. Previously published systematic reviews on assessment tools for NRS were not specifically focused on pharmacological interventions [9, 10], included randomized study designs [11-13], or investigated only a specific type of bias [14]. One systematic review of NRS tools for medications focused only on safety outcomes and it is now outdated, since published in 2012 [15]. Our systematic review has multiple strengths: authors reviewed the results of the searches independently following a pre-defined protocol; the framework for data extraction was developed based on inputs of worldwide experts in the field of pharmacoepidemiology and healthcare outcomes research coming from different backgrounds (academia, industry, governmental agency) and different countries; and it included the most updated versions of the identified tools. This review also has limitations. Search for tools in the grey

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literature might not be comprehensive since it was performed through only one browser. The search was also restricted to tools published in English and excluded identified tools that could not be retrieved.

Conclusion

In this systematic review, we found that available tools for NRS assessment failed to provide a comprehensive assessment of major methodological aspects that can affect the validity of NRS on the comparative safety and effectiveness of medications. Specifically, major aspects such as lack of new-user design, active comparator design, time-related bias (i.e., immortal time bias, time-window bias), and statistical assessment of internal validity, remain poorly covered. Including these critical elements into existing tools may provide a more accurate instrument to evaluate NRS of pharmacological interventions and increase awareness in the clinical research community about major addressable flaws in pharmacoepidemiology. This may improve the validity of NRS on the comparative safety and effectiveness of medications and reduce the publication of studies with unreliable findings.

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Conflict of Interest

Authors declare no conflict of interest

Author's Contribution

E.D. was involved in Substantial contributions to the conception and design, acquisition of data, analysis and interpretation of the data; drafting the article and revising it for intellectual content; and final approval of the version to be published.

L.V., G.S., T.P.A.D. were involved in Substantial contributions to the conception and design, acquisition of data, analysis and interpretation of the data; revising the article for intellectual content; and final approval of the version to be published.

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E.P., J.M.F. were involved in Substantial contributions to the conception and design, analysis and interpretation of the data; revising the article for intellectual content; and final approval of the version to be published.

D.B., J.A.L., D.C.M., H.Y., X.W., A.R.Z. were involved in Substantial contributions to the conception and design, interpretation of the data; revising the article for intellectual content; and final approval of the version to be published.

Ethics approval

Not needed. Since the systematic review did not involve primary data collection, and the consensus survey did not impose risks on participants, enrolling only competent adults, the protocol was not submitted for Institutional Review Board approval.

Transparency statement

The corresponding author affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned have been explained.

Patient and Public Involvement statement

This research was done without patient involvement. Patients were not invited to comment on the study design and were not consulted to develop patient relevant outcomes or interpret the results. Patients were not invited to contribute to the writing or editing of this document for readability or accuracy.

Data availability

All data relevant to the study are included in the article or uploaded as supplementary information.

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Table 1. Glossary of terms	
Term	**Definition**
Active comparator design	A study design that compares the effect of the drug of interest with another drug used in clinical practice instead of non-use.
Adjustment for causal intermediaries	Adjustment for an intermediate variable (or a descending proxy for an intermediate variable) on a causal path from exposure to outcome.
Case–control design	A study design in which cases (patients with outcomes) are identified and compared with controls (patients without outcomes) with respect to the exposure of interest.
Cohort design	A study design in which a group of patients (a cohort) is identified and followed to ascertain the occurrence of an outcome.
Confounding	A mixing of effects that arises when patients with different baseline risks are compared; the resulting effect measure is a mix of drug effects and risk factor effects.
Depletion of susceptibles	Selection bias that occurs when the initiation of exposure to a drug is associated with an early increased incidence rate of the study outcome, followed by a decreased incidence rate with longer duration of exposure (e.g., users of new drugs are compared to users of older drugs).
Detection or surveillance bias	Bias that occurs when the degree of outcome surveillance (or an associated symptom) is related to exposure and is differential among the exposure groups.
Immortal time bias	Time-related bias that derives from including a period of follow-up during which, by design, outcomes cannot occur.
Time-window bias	Time-related bias, in the context of a case-control study nested in a cohort, that derives from the use of time-windows of different lengths between cases and controls to define time-dependent exposures.
Incorrect outcome model specification	Misspecification of a statistical model that leads to biased outcome results. Common causes are omission of a relevant variable, inclusion of an unnecessary variable, adopting the wrong functional form, incorrect specification of the error term, uncertainty about what the true model is, reciprocal causation.
Loss to follow up bias	Bias that occurs when there is difference in retention during the follow-up period after enrollment that are related to exposure status and outcome.
New-user design	A study design that starts following patients at the time they initiate a new drug (also known as incident-user design)
Non-contemporaneous comparator bias	Bias generated by differences in the timing of selection of comparator group(s) within a study influence exposures and outcomes resulting in biased estimates.
Reverse causation (or reverse causality)	Bias due to direction of cause-and-effect contrary to a common presumption, or a two-way causal relationship between exposure and outcome.
Recall bias	Bias that occurs when participants do not remember previous events or experiences accurately or omit details (not for claims-based studies).
Selection bias	Bias that occurs when selection of participants or follow-up time is related to both intervention and outcome (e.g. prevalent users of a drug are compared with non-users or incident users). Our framework has a separate sub-domain that refers to selection bias due to lack of generalizability, applicability or transferability to patients who were excluded from the study.

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Table 2. Individual characteristics of the tools included in the systematic review

Tool identified*	Year	Type of tool	Scope of the tool	Study design evaluated	Items
RELEVANT	2019	Checklist	Critical appraisal and reporting	NRS	21
RAMboMAN - GATE-EPIQ	2019	Rating scale + summary judgment	Critical appraisal	Coh (+RCTs), CC	Coh (+RCTs) 21, CC 18
MMAT	2018	Checklist	Critical appraisal	NRS	5
CASP	2018	Checklist	Critical appraisal	Coh, CC	Coh 12, CC 11
SURE	2018	Checklist + summary judgment	Critical appraisal	Coh, CC	Coh 13, CC 11
JBI	2017	Checklist + summary judgment	Critical appraisal	Coh, CC	Coh 11, CC 10
ROBINS-I	2016	Checklist + summary judgment	Critical appraisal	NRS	34 (+ 8 optional question)
ISPOR-AMCP-NPC (#)	2014	Checklist + summary judgment	Critical appraisal	Coh CC	32
GRACE (#)	2014	Checklist + summary judgment	Critical appraisal	Coh CC	11
NIH–NHLBI	2014	Checklist + summary judgment	Critical appraisal	Coh (+ CSS), CC	Coh (+ CSS) 14, CC 12
HEBW	2014	Checklist + summary judgment	Critical appraisal	Coh	18
RoBANS	2013	Rating scale	Critical appraisal	NRS	6
RTI-Item Bank	2013	Checklist	Critical appraisal	NRS	13
Newcastle-Ottawa	2013	Rating scale + summary judgment	Critical appraisal	Coh, CC	Coh 8, CC 8
SIGN - Version 3.0	2012	Checklist + summary judgment	Critical appraisal	Coh, CC	Coh 14, CC 11
Montreal	2011	Checklist	Critical appraisal	Coh CC (+RCTs)	10
EPHPP	2011	Rating scale	Critical appraisal	Coh CC (+RCTs)	17
STROBE - Version 4	2007	Checklist	Reporting	Coh, CC	Coh 22, CC 22
TREND	2004	Checklist	Reporting	NRS	22
Margetts	2002	Checklist	Reporting	Coh CC	11
Zaza	2000	Checklist	Critical appraisal	Coh CC	15
Downs-Black	1998	Rating scale	Critical appraisal and reporting	Coh CC (+RCTs)	27
Elwood	1998	Checklist	Critical appraisal	Coh CC (+RCTs)	20
Hadorn	1996	Checklist	Critical appraisal	Coh (+RCTs)	7
London	1996	Checklist	Critical appraisal	Coh CC	33
Avis	1994	Rating scale + summary judgment	Critical appraisal and reporting	Coh CC (+RCTs)	24
Durant	1994	Checklist	Critical appraisal	CC	23
Levine	1994	Checklist	Critical appraisal	Coh CC (+RCTs)	10
Gyorkos	1994	Checklist	Critical appraisal	Coh, CC	Coh 6, CC 5
Cho (#)	1994	Rating scale + summary judgment	Critical appraisal	NRS (+RCTs)	24
COEH	1991	Checklist	Critical appraisal	NRS	54
Fowkes-Fulton	1991	Checklist + summary judgment	Critical appraisal and reporting	NRS (+RCTs)	6
Lichtenstein	1987	Checklist	Critical appraisal	CC	20
Gardner	1986	Checklist	Critical appraisal	NRS	12
Horwitz	1979	Checklist	Critical appraisal and reporting	CC	12

Notes: Nine records from our bibliographic search provided two separate tools to assess cohort or case-control studies. Thus, the overall number of included records is 34, while the number of included assessment tools is 43.
NRS = non-randomized study, Coh = cohort study, CC = case control, RCT = randomized controlled trial, CSS = cross-sectional study, NRS = Non-randomized studies.
* tool name or first author name, if the tool does not have an assigned name and it was published in peer-review journals.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.
Acronym of the tools: RELEVANT = The REal Life EVidence AssessmeNt Tool; RAMboMAN, GATE-EPIQ = Recruitment Allocation Maintenance blind objective Measurements Analyses, Graphic Approach To Epidemiology - Effective Practice, Informatics and Quality Improvement; MMAT = Mixed Methods Appraisal Tool; CASP = The Critical Appraisals Skills Programme; SURE Specialist Unit for Review Evidence; JBI = The Joanna Briggs Institute; ROBINS-I = Risk Of Bias In Non-randomized Studies - of Interventions; ISPOR-AMCP-NPC =

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International Society for Pharmacoeconomics and Outcomes Research - Academy of Managed Care Pharmacy - National Pharmaceutical Council; GRACE = The Good ReseArch for Comparative Effectiveness; NIH–NHLBI = The National Institute of Health - The National Heart, Lung, and Blood Institute; HEBW = Health Evidence Bulletins Wales; RoBANS = Risk of Bias Assessment tool for Non-randomized Studies; RTI-Item Bank = Research Triangle Institute Item Bank; SIGN = The Scottish Intercollegiate Guidelines Network; STROBE = STrengthening the Reporting of OBservational studies in Epidemiology; EPHPP = Effective Public Health Practice Project Quality Assessment Tool; TREND = Transparent Reporting of Evaluations with Nonrandomized designs; COEH = Centre for Occupational and Environmental Health of The University of Manchester.

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Table 3. General characteristics of the tools included in the systematic review				
Characteristics	**All, n = 44**	**Cohort *, N = 11**	**Case-control, n = 12**	**NRS**, n = 21**
Publication year – n (%)				
1979-1989	3 (7)	0 (0)	2 (17)	1 (5)
1990-1999	12 (27)	2 (18)	2 (17)	8 (38)
2000-2009	5 (11)	1 (9)	1 (8)	3 (14)
2010-2019	24 (55)	8 (73)	7 (58)	9 (43)
Type of tool – n (%)				
Checklist	22 (50)	4 (36)	6 (50)	12 (57)
Checklist + summary judgment	13 (30)	5 (45)	4 (33)	4 (19)
Rating scale	3 (7)	0 (0)	0 (0)	3 (14)
Rating scale + summary judgment	6 (14)	2 (18)	2 (16)	2 (9)
Scope of the tool – n (%)				
Critical appraisal	35 (80)	9 (81)	10 (83)	16 (76)
Reporting	4 (9)	2 (18)	1 (8)	1 (5)
Critical appraisal and Reporting	5 (11)	0 (0)	1 (8)	4 (19)
Tools designed for CER – n (%)	3 (7)	0 (0)	0 (0)	3 (14)
Number of items – median (IQR)	13 (10.3-21.8)	13 (9.5-16)	11.5 (10.8-18.5)	17 (11-24)

CER = Comparative Effectiveness research
* two tools evaluated both cohort and RCTs together, one tool evaluated both cohort and cross-sectional studies together
** NRS tools refer to a single tool built to evaluate both cohort and case-control studies, or a tool built to evaluate additional NRS (e.g. cross-sectional studies, before-after studies) together with cohort and case-control studies. Eight NRS tools included also the evaluation of RCTs

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Table 4. Methodological challenges addressed by the included tools

Domains		Cohort tools*, N = 11	Case-control tools, n = 12	NRS tools**, n = 21	Total, n = 44
1. Methods for selecting participants – n (%)		11 (100)	12 (100)	20 (95)	43 (98)
	Sampling strategies to correct selection bias	4 (36)	6 (50)	9 (42)	19 (43)
	Inclusion and exclusion criteria of target population	6 (55)	8 (67)	13 (61)	27 (61)
	Depletion of susceptibles	1 (9)	0 (0)	0 (0)	1 (2)
	External validity of target population	6 (55)	6 (50)	9 (43)	21 (48)
	Others***	11 (100)	12 (100)	18 (86)	41 (93)
2. Measurement of exposure, outcomes, covariates, and follow-up – n (%) ^		11 (100)	12 (100)	19 (90)	42 (95)
	Measurement of exposure ^	11 (100)	11 (92)	18 (81)	40 (91)
	Measurement of outcomes ^	11 (100)	11 (92)	18 (81)	40 (91)
	Measurement of covariates	4 (36)	4 (33)	4 (19)	12 (27)
	Measurement of follow-up	9 (82)	3 (25)	5 (24)	17 (39)
3. Design-specific sources of bias – n (%)		11 (100)	10 (83)	19 (90)	40 (91)
	New user design	0 (0)	0 (0)	2 (10)	2 (5)
	Active comparator design	0 (0)	0 (0)	0 (0)	0 (0)
	Immortal time bias or time-window bias	0 (0)	0 (0)	3 (14)	3 (7)
	Detection or surveillance bias	1 (9)	2 (17)	1 (5)	4 (9)
	Loss to follow-up bias	9 (82)	1 (8)	12 (57)	22 (50)
	Non-contemporaneous comparator bias	0 (0)	1 (8)	5 (24)	6 (14)
	Reverse causation	5 (45)	1 (8)	2 (10)	8 (18)
	Recall bias^^	1 (9)	4 (33)	1 (5)	6 (14)
	Interviewer or observer bias^^	1 (9)	3 (25)	7 (35)	11 (25)
	Ascertainment bias^^	0 (0)	1 (8)	1 (5)	2 (5)
	General item/question on bias^^	3 (27)	3 (25)	3 (14)	9 (20)
	Other biases^^^	0 (0)	2 (17)	5 (24)	7 (16)
4. Confounding – n (%)		11 (100)	11 (92)	18 (86)	40 (91)
	Study design used to minimize confounding	6 (55)	7 (58)	13 (62)	26 (59)
	Confounders measured and included in statistical analyses	10 (91)	10 (83)	18 (86)	38 (86)
	Potential unmeasured confounding addressed in the analysis (e.g., proxy analysis, IV analysis)	1 (9)	1 (8)	3 (14)	5 (11)
5. Lack of appropriateness of statistical analyses (with specific mention of overadjustment, and/or incorrect outcome model specification) – n (%)		2 (18)	3 (25)	10 (48)	15 (34)
6. Methods for assessing statistical uncertainty in the findings (e.g., confidence intervals reported for each analysis) – n (%)		7 (64)	6 (50)	8 (38)	21 (48)
7. Methods for assessing internal validity (e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases) – n (%)		3 (27)	3 (25)	9 (43)	15 (34)
8. Methods for assessing external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.) – n (%)		4 (36)	3 (25)	4 (19)	11 (25)

* two tools evaluated both cohort and RCTs together, one tool evaluated both cohort and cross-sectional studies together

** NRS tools refer to a single tool built to evaluate both cohort and case-control studies, or a tool built to evaluate additional NRS (e.g. cross-sectional studies, before-after studies) together with cohort and case-control studies. Eight NRS tools included also the evaluation of RCTs

*** "others" refers to items not included in our evaluation framework but included in the reviewed tools to investigate selection bias (e.g., population characteristics sufficiently described to determine the applicability of the research question, sample size justification and power description, ethical considerations etc.)

^ items or questions on exposure misclassification and/or outcome misclassification are counted in this domain and relative subdomains

^^ design-specific biases not included in the evaluation framework but addressed by the reviewed tools

^^^ other design-specific biases not included in the evaluation framework but addressed by a few tools (e.g., bias due to missing data, patients' blinding, different length of follow-up between groups, Berkson's bias, protopathic bias) ^

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Supplement file 1

Table S1. Search strategy

Table S2. List of investigated websites of influential initiatives, programs or organizations.

Figure S1. Flowchart of the selection of the tools

Figure S2. Flowchart of the Delphi procedure

Table S3. Excluded tools after full-text screening

Figure S3. Assessment tools that included or not at least one question/item to address subdomains of methods for selecting study participants (Domain 1), stratified by study design.

Figure S4. Assessment tools that included or not at least one question/item to address subdomains for measurement of exposure, outcomes, covariate variables and follow-up (Domain 2), stratified by study design.

Figure S5. Assessment tools that included or not at least one question/item to address subdomains to detect design-specific sources of biases (Domain 3), stratified by study design.

Figure S6. Assessment tools that included or not at least one question/item to address confounding (Domain 4), stratified by study design**.**

Figure S7. Assessment tools that included or not at least one question/item to address lack of appropriateness of statistical analyses (Domain 5), methods for assessing statistical uncertainty in the findings (Domain 6), methods for assessing internal validity (Domain 7), methods for assessing external validity (Domain 8), stratified by study design.

Figure S8. Proportion of domains covered by each reviewed tool

Figure S9. Proportion of domains covered by each reviewed tool of case-control studies only

Figure S10. Proportion of domains covered by each reviewed tool of cohort studies only

Figure S11. Proportion of domains covered by each reviewed tool of multiple study designs (NRS)

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In this supplement, we describe in greater detail the methods and results of our review process and Delphi survey and report additional findings related to the evaluation of each assessment tool.

As stated in the main manuscript, the framework to evaluate the assessment tools was developed based on the domains and subdomains indicated by the Comparative Effectiveness Research (CER) Special Interest Group (SIG) as the main critical elements that can threat to the validity of non-randomized studies (NRS) of medications.

Delphi Survey

To collect the domains and to ultimately develop the evaluation framework, we performed a survey following the Delphi methodology. This is an iterative multistage process that uses rounds of questionnaires to reach a consensus among content experts [Hasson et al. 2000]. The survey targeted all members of the SIG of the International Society for Pharmacoepidemiology CER. These experts were informed about the survey during an online meeting hold on July 1, 2019. During the meeting, study protocol and Delphi survey were presented to the group, addressing that answering to the survey would have constituted consent to participate in the study. After the online meeting, the study protocol was e-mailed to the members of CER SIG.

On August 23, 2019, a link to a self-administered online questionnaire was sent by email to 48 CER SIG members. In the first page of the survey, a cover letter outlined the purpose of the study and assured the participants of anonymity in processing the results. The online questionnaire could have only been submitted once per person. The experts had three weeks to answer after the initial e-mail, during this time one remainder was sent.

The two-section questionnaire was developed through several face-to-face and online meetings among the authors. The first section included questions on demographic and professional information of participants. In the second section, respondents were asked to rate, on a five-point scale ranging from 1 (“not important”) to 5 (“extremely important”), which items should be considered when assessing the overall quality of a NRS, specifically cohort and case-control studies, of pharmacological interventions.

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In addition to the rating, the respondents had the option of providing any other relevant considerations at the end of the questionnaire (see Supplement 2).

The outcome variable of the online survey, originally consisting of five categories, was collapsed into two levels. Participants considered important an item if they rated it 4 or 5 (where 5 is "extremely important"), while they did not consider an item of key importance if they rated it 1, 2 or 3 (where 1 is "not important"). When at least 70% of participants agreed on the importance of an item (rating 1, 2 or 3 vs rating 4 or 5), we considered that a consensus had been reached on that item.

Thirty-five CER SIG experts participated to the first round of the survey (response rate = 73%). Eighteen respondents worked in academia (51%), 16 respondents worked in industry (46%), and only one was affiliated to a governmental institution (3%). The majority of respondents were from U.S. (n = 26, 74%), the others were from Europe (n = 7, 20%) or from Canada (n = 2, 6%). After participants ranked the relevance of the items and provided comments to the first questionnaire, we reviewed the responses and adjusted the survey to reflect the feedback. Overall, participants considered the presented domains important or extremely important. All domains, other than the domain n. 5 ("Lack of appropriateness of statistical analyses"), were rated relevant with unanimous consent (100% of participants rated them 4 or 5). For the domain n. 5 ("Lack of appropriateness of statistical analyses (with specific mention of adjustment for causal intermediaries, and/or incorrect outcome model specification)" in the Supplement 2), only three participants did not consider it important. Based on individual experts' suggestions, we added the subdomain "External validity of target population" to domain n. 1 "Methods of selecting participants" to address the external validity or generalizability of the study population to the eligible or target population. We removed "Recall bias" from the critical elements of domain n. 3 since this was the only domain to be rated not relevant by most of participants (n = 26, 74%). The rationale provided was that, since real world data (RWD) are often collected for intents unrelated to pursue research goals (e.g., administrative purposes), biases such as recall bias, interviewer bias, nonresponse bias are usually reduced or even eliminated [Sørensen HT et al.].

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A revised survey was returned to the same participants for reassessment of the content. The response rate to the second round was 89% (n = 31/35). Since in this second round the threshold of 70% was reached for all the old and new items, we terminated the consensus procedure and communicated the results to the CER SIG members. The Delphi flowchart is shown in Figure S2, while the last version of cover letter and online questionnaire is available on the online Supplement 2.

The evaluation framework was built on the main domains and subdomains identified through the Delphi process and it was used as an instrument to evaluate the completeness of the existing assessment tools for cohort and case-control studies only or multiple NRS. The assessment of each domain and subdomain and the distribution of domains in the reviewed tools are shown in Figures S3-S11.

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Table S1. Search strategy

Items	N.	Terms
Critical appraisal tool	#1	"critical" [All Fields] AND "appraisal" [All Fields] AND "tools" [All Fields]
	#2	"critical" [All Fields] AND "appraisal" [All Fields]
	#3	("critical" [All Fields] AND "review" [All Fields]) OR "critical review" [All Fields]) AND form [All Fields]
	#4	("systematic review" [Publication Type] OR "systematic reviews as topic" [MeSH Terms] OR "systematic review"[All Fields]) AND form [All Fields]
	#5	appraisal [All Fields] AND ("research design" [MeSH Terms] OR ("research" [All Fields] AND "design" [All Fields]) OR "research design" [All Fields] OR ("research" [All Fields] AND "methodology" [All Fields]) OR "research methodology"[All Fields])
	#6	("research design" [MeSH Terms] OR ("research" [All Fields] AND "design" [All Fields]) OR "research design" [All Fields]) AND ("review" [Publication Type] OR "review literature as topic" [MeSH Terms] OR "review"[All Fields])
Study reporting tool	#7	"study" [All Fields] AND "reporting" [All Fields] AND "tool" [All Fields]
	#8	"study" [All Fields] AND "reporting" [All Fields]
	#9	"reporting" [All Fields] AND "form" [All Fields] AND ("Studies"[Journal] OR "studies"[All Fields])
	#10	"reporting" [All Fields] AND ("Studies"[Journal] OR "studies"[All Fields])
Tool	#11	"checklist" [MeSH Major Topic] OR "scale*" [Title/Abstract]
	#12	"surveys and questionnaires"[MeSH Major Topic] OR "questionnaire*" [Title/Abstract]
	#13	("tool*" [All Fields] OR "instrument*" [All Fields] OR "checklist*" [All Fields] OR "questionnaire*" [All Fields]) AND ("quality" [All Fields] OR "method*" [All Fields] OR "bias" [All Fields])
Study design	#14	"cohort studies"[MeSH Terms] OR cohort studies [Text Word] OR cohort stud* [All Fields]
	#15	"case-control studies" [MeSH Terms] OR case-control studies [Text Word] OR case control stud* [All Fields]
	#16	Non [All Fields] AND ("random allocation"[MeSH Terms] OR randomized [Text Word]) AND stud* [All Fields]
Systematic review	#17	"systematic review" [Publication Type] OR "systematic reviews as topic"[MeSH Terms] OR "systematic review"[All Fields]
Filters	#18	"humans"[MeSH Terms]
	#19	"Review" [ptyp] OR "systematic" [sb]
Strings		
1st search - tools*	#20	(#1 OR #2 OR #3 OR #4 OR #5 OR #6) AND (#14 OR #15 OR #16) AND #18
	#21	(#7 OR #8 OR #9 OR #10) AND (#14 OR #15 OR #16) AND #18
	#22	(#11 OR #12 OR #13) AND (#14 OR #15 OR #16) AND #18
2nd search - systematic reviews of tools*	#23	(#1 OR #2 OR #3 OR #4 OR #5 OR #6) AND (#14 OR #15 OR #16) AND #18 AND (#17 OR #19)
	#24	(#7 OR #8 OR #9 OR #10) AND (#14 OR #15 OR #16) AND #18 AND (#17 OR #19)
	#25	(#11 OR #12 OR #13) AND (#14 OR #15 OR #16) AND #18 AND (#17 OR #19)

*tools are defined as instruments (e.g., qualitative checklists, questionnaires, scoring scales, etc.) that investigate the overall quality of a study, identifying potential biases – either used to critically appraise studies included in a systematic review, or to help in the peer-reviewing process of scientific publications – (i.e., critical appraisal tools), or that support the reporting of research methods and findings (i.e. study reporting tools).
Note: the strings were built in Medline and then adapted to Embase (through Elsevier)

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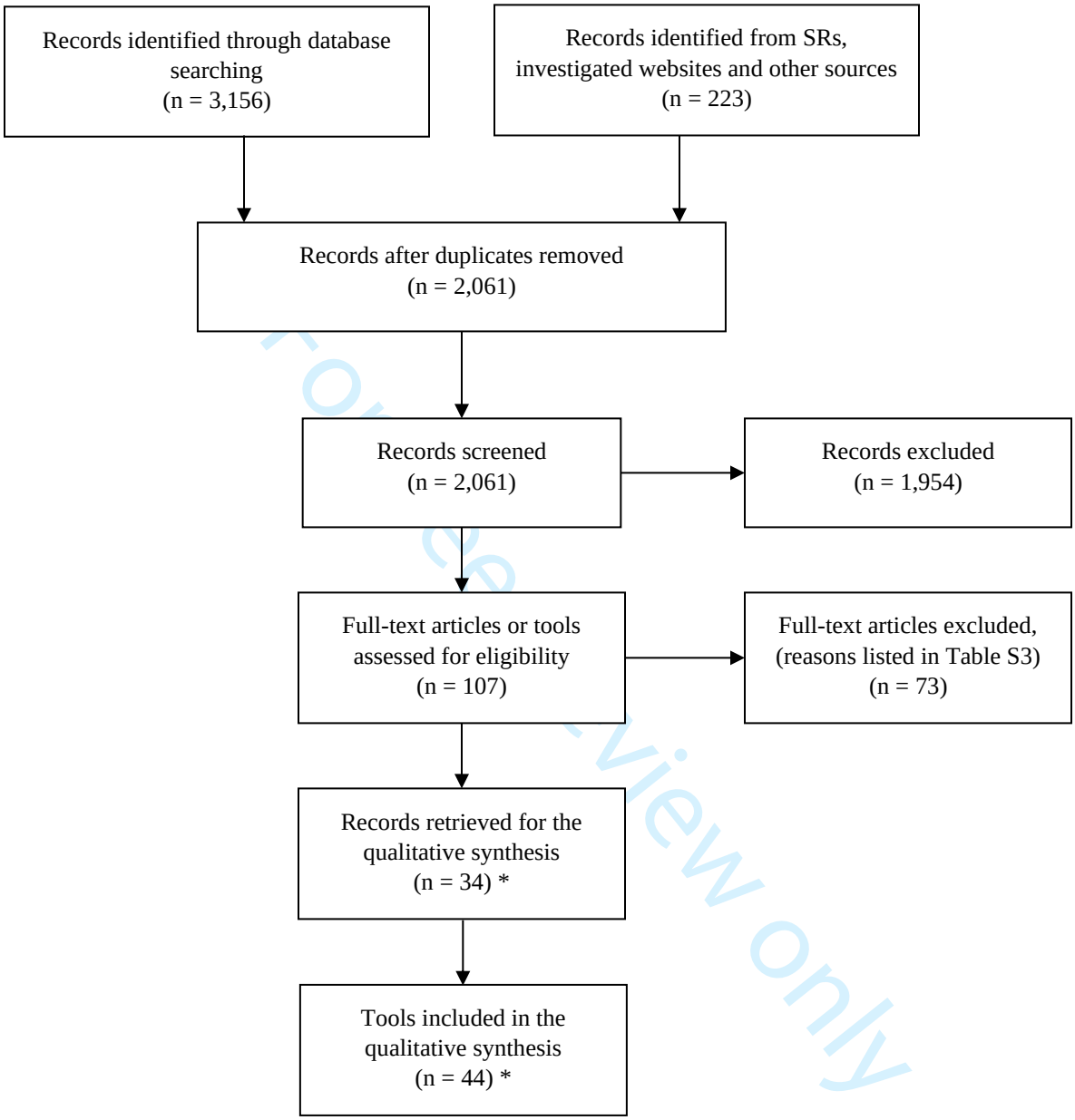
Table S2. List of investigated websites of influential initiatives, programs or organizations.

Name of initiative, programs or institutions	Link to the website
STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) statement	https://strobe-statement.org/index.php?id=available-checklists
Effective Health Care (EHC) Program of the Agency for Healthcare Research and Quality (AHRQ)	https://effectivehealthcare.ahrq.gov/research-methods
The Ottawa Hospital – Research Institute	http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp
The Scottish Intercollegiate Guidelines Network (SIGN)	https://www.sign.ac.uk/checklists-and-notes.html
GRACE (Good ReseArch for Comparative Effectiveness) initiative	https://www.graceprinciples.org/about-grace.html
The Joanna Briggs Institute	http://joannabriggs.org/research/critical-appraisal-tools.html
TREND (Transparent Reporting of Evaluations with Nonrandomized Designs) statement	https://www.cdc.gov/trendstatement/index.html
CASP (Critical Appraisal Skills Programme)	https://casp-uk.net/casp-tools-checklists/

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Figure S1. Flowchart of the selection of the tools



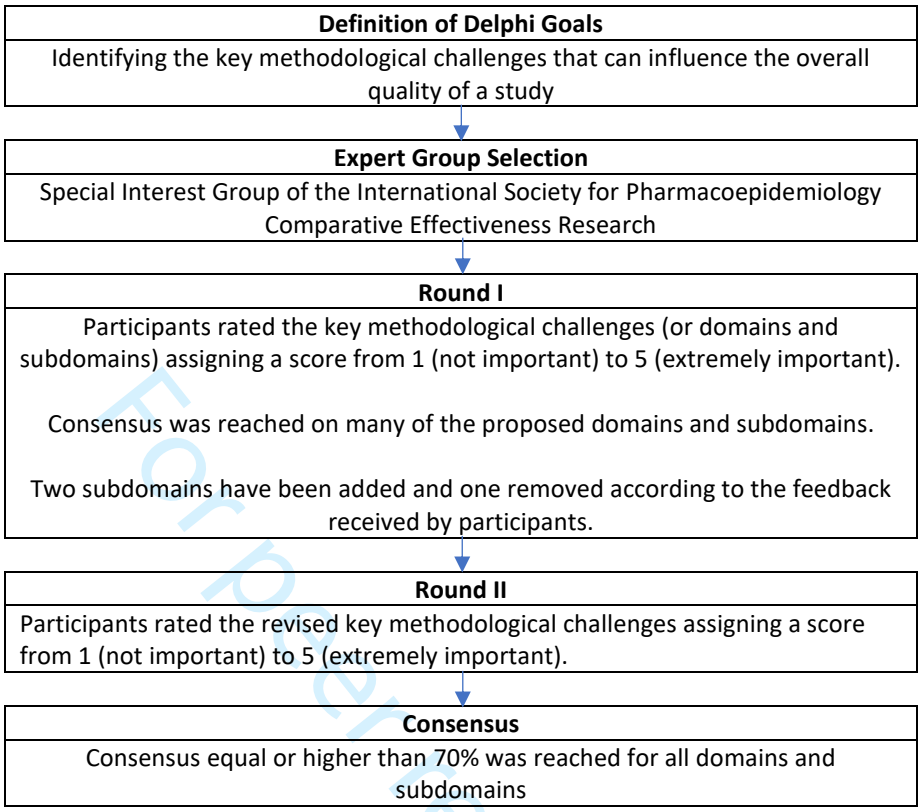
*Ten retrieved articles/records included two different assessment tools for case-control and cohort studies. Thus, the overall number of included tools raised up to 44.

The systematic reviews (SRs) were retrieved through a systematic search. Additional details about the SRs search are available upon request.

NB. The reporting items in the flowchart follows the indications given by the PRISMA Statement at Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097.

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Figure S2. Flowchart of the Delphi procedure



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Table S3. Excluded tools after full-text screening

Reference (73 article excluded)	**Reason for exclusion**
Berkman ND, Lohr KN, Ansari M, et al. Chapter 15 Appendix A: A Tool for Evaluating the Risk of Reporting Bias (in Chapter 15: Grading the Strength of a Body of Evidence When Assessing Health Care Interventions for the Effective Health Care Program of the Agency for Healthcare Research and Quality: An Update). Methods Guide for Comparative Effectiveness Reviews (Prepared by the RTI-UNC Evidence-based Practice Center under Contract No. 290-2007-10056-I). AHRQ Publication No. 13(14)-EHC130-EF. Rockville, MD: Agency for Healthcare Research and Quality, 2013	Assessment of systematic reviews
Balshem H, Stevens A, Ansari M, et al. Finding grey literature evidence and assessing for outcome and analysis reporting biases when comparing medical interventions: AHRQ and the Effective Health Care Program. (Prepared by the Oregon Health and Science University and the University of Ottawa Evidence-based Practice Centers under Contract Nos. 290-2007-10057-I and 290-2007-10059-I.) AHRQ Publication No. 13(14)-EHC096-EF. Rockville, MD: Agency for Healthcare Research and Quality, 2013	Assessment of reporting bias, guidance
Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 2008; 336:924–6.	Assessment to provide strength of recommendations
Hayden JA van der Windt DA, Cartwright JL, et al. Assessing bias in studies of prognostic factors. Ann Intern Med 2013; 158:280–6.	Assessment of studies of prognostic factors
Higgins JPT, Altman DG, Sterne JAC. Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S, eds. Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0. London: The Cochrane Collaboration, 2011.	Assessment of RCTs
Higgins JPT, Savović J, Page MJ, et al. Revised Cochrane risk of bias tool for randomized trials (RoB 2.0), Version 20. 2016 Available at: https://sites. google.com/site/riskofbiastool/ (accessed 19 Sep 2019)	Assessment of RCTs
Hooijmans CR, Rovers MM, de Vries RB, et al. SYRCLE's risk of bias tool for animal studies. BMC Med Res Methodol 2014; 14:43	Assessment of animal studies
Slim K, Nini E, Forestier D, Kwiatkowski F, Panis Y, Chipponi J. Methodological index for non-randomized studies (MINORS): development and validation of a new instrument. ANZ J Surg. 2003;73(9): 712-716.	Assessment of NRS of surgical interventions
Centre for reviews and dissemination. Systematic Reviews: CRD's Guidance for Undertaking Reviews in Healthcare. 3rd ed. UK: University of York; 2009.	No assessment tool, guidance for reviews
Deeks JJ, Dinnes J, D'Amico R, et al. Evaluating non-randomised intervention studies. Health Technol. Assess (Winchester, England. 2003;7(27): iii-x):1-173.	No assessment tool, original research
NICE. Guide to the methods of technology appraisal 2013. Updated 2016.	No assessment tool, guidance for HTA
University of Wales College of Medicine. Critical Appraisal Forms. University of Wales, 2004.	N/A
Loney PL, Chambers LW, Bennett KJ, Roberts JG, Stratford PW. Critical appraisal of the health research literature: prevalence or incidence of a health problem. Chronic Dis Canada 2000;19:170–77.	No structured assessment tool
Cameron I, Crotty M, Currie C et al. Geriatric rehabilitation following fractures in older people: a systematic review. Health Technol Assess 2000;4: i–111.	No assessment tool
Carneiro AV. Critical appraisal of prognostic evidence: practical rules. Rev Port Cardiol 2002; 21:891–900.	Assessment tool of prognostic studies
Centre for Evidence-Based Mental Health, University of Oxford. Critical Appraisal Forms. 2004. Available at: http://www.cebmh.com/ (accessed 19 Sep 2019)	Assessment tool tailored to a particular disease/topic

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Esdaile JM, Horwitz RI. Observational studies of cause-effect relationships: an analysis of methodologic problems as illustrated by the conflicting data for the role of oral contraceptives in the etiology of rheumatoid arthritis. J Chronic Dis 1986; 39:841–52.	No structured assessment tool
Khan KS, Riet GT, Popay J, Nixon J, Kleijnen J. Undertaking systematic reviews of research effectiveness. CRD's guidance for those carrying out or commissioning reviews. CRD Report number 4 (2nd edn). 2001. The University of York Centre for Reviews and Dissemination.	No structured assessment tool
How to read clinical journals: IV. To determine etiology or causation. Can Med Assoc J. 1981;124(8):985-990.	No structured assessment tool for NRS
Mulrow CD, Lichtenstein MJ. Blood glucose and diabetic retinopathy: a critical appraisal of new evidence. J Gen Intern Med 1986; 1:73–77.	Assessment tool tailored to a particular disease/topic
Whiting P, Rutjes AW, Reitsma JB, Bossuyt PM, Kleijnen J. The development of QUADAS: a tool for the quality assessment of studies of diagnostic accuracy included in systematic reviews. BMC Med Res Methodol 2003; 3:25.	Assessment tool for diagnostic accuracy studies
Campbell H, Rudan I. Interpretation of genetic association studies incomplex disease. Pharmacogenomics J 2002; 2:349–60.	Assessment tool tailored to a particular disease/topic
Solomon DH, Bates DW, Panush RS, Katz JN. Costs, outcomes, and patient satisfaction by provider type for patients with rheumatic and musculoskeletal conditions: a critical review of the literature and proposed methodologic standards. Ann Intern Med 1997; 127:52–6	Assessment tool tailored to a particular disease/topic
The STARD Group. The STARD Initiative—Towards Complete and Accurate Reporting of Studies on Diagnostic Accuracy. http://www.consort-statement.org/stardstatement.htm, 2001.	Assessment tool for diagnostic accuracy studies
Critical appraisal: Guidelines for the critical appraisal of a paper. Surgical-Tutor.org.uk, 2004.	N/A
Zola P, Volpe T, Castelli G et al. Is the published literature a reliable guide for deciding between alternative treatments for patients with early cervical cancer? Int J Radiat Oncol Biol Phys 1989;16:785–97.	No assessment tool, original research
Bollini P, Garcia Rodriguez LA, Gutthann SP, Walker AM. The impact of research quality and study design on epidemiologic estimates of the effect of nonsteroidal anti-inflammatory drugs on upper gastrointestinal tract disease. Arch Intern Med 1992; 152:1289–95.	No assessment tool, original research
Ciliska D, Hayward S, Thomas H et al. A systematic overview of the effectiveness of home visiting as a delivery strategy for public health nursing interventions. Can J Public Health 1996;87:193–98.	No assessment tool, original research
Cowley DE. Prostheses for primary total hip replacement. A critical appraisal of the literature. Int J Technol Assess Health Care 1995;11:770–78.	No assessment tool, original research
Spitzer WO, Lawrence V, Dales R et al. Links between passive smoking and disease: a best-evidence synthesis. A report of the Working Group on Passive Smoking. Clin Invest Med 1990;13: 17–42.	No assessment tool, original research
Steinberg EP, Eknoyan G, Levin NW et al. Methods used to evaluate the quality of evidence underlying the National Kidney Foundation- Dialysis Outcomes Quality Initiative Clinical Practice Guidelines: description, findings, and implications. Am J Kidney Dis 2000; 36:1–11.	No assessment tool, original research
Anders JF, Jacobson RM, Poland GA, Jacobsen SJ, Wollan PC. Secondary failure rates of measles vaccines: a meta-analysis of published studies. Pediatr Infect Dis J 1996; 15:62–66.	No assessment tool, original research
Ariens GA, van Mechelen W, Bongers PM, Bouter LM, van der WG. Physical risk factors for neck pain. Scand J Work Environ Health 2000; 26:7–19.	No assessment tool, original research
Berlin JA, Colditz GA. A meta-analysis of physical activity in the prevention of coronary heart disease. Am J Epidemiol 1990; 132:612–28.	No assessment tool, original research
Bhutta AT, Cleves MA, Casey PH, Cradock MM, Anand KJS. Cognitive and behavioral outcomes of school-aged children who were born preterm: a meta-analysis. J Am Med Assoc 2002;288: 728–37.	No assessment tool, original research

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Borghouts JA, Koes BW, Bouter LM. The clinical course and prognostic factors of non-specific neck pain: a systematic review. Pain 1998; 77:1–13.	No assessment tool, original research
Campos-Outcalt D, Senf J, Watkins AJ, Bastacky S. The effects of medical school curricula, faculty role models, and biomedical research support on choice of generalist physician careers: a review and quality assessment of the literature. Acad Med 1995; 70:611–19.	No assessment tool, original research
Carson CA, Fine MJ, Smith MA, Weissfeld LA, Huber JT, Kapoor WN. Quality of published reports of the prognosis of community-acquired pneumonia. J Gen Intern Med 1994; 9:13–19.	No assessment tool, original research
Corrao G, Bagnardi V, Zambon A, Arico S. Exploring the doseresponse relationship between alcohol consumption and the risk of several alcohol-related conditions: a meta-analysis. Addiction 1999; 94:1551–73.	No assessment tool, original research
Garber BG, Hebert PC, Yelle JD, Hodder RV, McGowan J. Adult respiratory distress syndrome: a systemic overview of incidence and risk factors. Crit Care Med 1996; 24:687–95.	No assessment tool, original research
Goodman SN, Berlin J, Fletcher SW, Fletcher RH. Manuscript quality before and after peer review and editing at Annals of Internal Medicine. Ann Intern Med 1994; 121:11–21.	No assessment tool
Jabbour M, Osmond MH, Klassen TP. Life support courses: are they effective? Ann Emerg Med 1996; 28:690–98.	No assessment tool, original research
Kreulen CM, Creugers NH, Meijering AC. Meta-analysis of anterior veneer restorations in clinical studies. J Dent 1998; 26:345–53.	No assessment tool, original research
Krogh CL. A checklist system for critical review of medical literature. Med Educ 1985; 19:392–95.	No assessment tool for NRS validity
Littenberg B, Weinstein LP, McCarren M et al. Closed fractures of the tibial shaft. A meta-analysis of three methods of treatment. J Bone Joint Surg Am 1998; 80:174–83.	No assessment tool, original research
Longnecker MP, Berlin JA, Orza MJ, Chalmers TC. A meta-analysis of alcohol consumption in relation to risk of breast cancer. JAMA 1988; 260:652–56.	No assessment tool, original research
Macfarlane TV, Glenny AM, Worthington HV. Systematic review of population-based epidemiological studies of oro-facial pain. J Dent 2001;29:451–67.	No assessment tool, original research
Manchikanti L, Singh V, Vilims BD, Hansen HC, Schultz DM, Kloth DS. Medial branch neurotomy in management of chronic spinal pain: systematic review of the evidence. Pain Physician 2002;5:405–18.	No assessment tool, original research
Meijer R, Ihnenfeldt DS, van Limbeek J, Vermeulen M, de Haan RJ. Prognostic factors in the subacute phase after stroke for the future residence after six months to one year. A systematic review of the literature. Clin Rehabil 2003;17:512–20.	No assessment tool, original research
Nguyen QV, Bezemer PD, Habets L, Prahl-Andersen B. A systematic review of the relationship between overjet size and traumatic dental injuries. Eur J Orthod 1999;21:503–15.	No assessment tool, original research
Rangel SJ, Kelsey J, Colby CE, Anderson J, Moss RL. Development of a quality assessment scale for retrospective clinical studies in pediatric surgery. J Pediatr Surg 2003;38:390–96.	Assessment tool tailored to a particular disease/topic
Reisch JS, Tyson JE, Mize SG. Aid to the evaluation of therapeutic studies. Pediatrics 1989;84:815–27.	No assessment tool
Stock SR. Workplace ergonomic factors and the development of musculoskeletal disorders of the neck and upper limbs: a metaanalysis. Am J Ind Med 1991;19:87–107.	No assessment tool, original research
van der Windt DAWM, Thomas E, Pope DP et al. Occupational risk factors for shoulder pain: a systematic review. Occup Environ Med 2000;57:433–42.	No assessment tool, original research
Margetts BM, Vorster HH, Venter CS. Evidence-based nutrition—review of nutritional epidemiological studies. South African J Clin Nutr 2002;15:68–73.	Assessment tool tailored to a particular disease/topic
Higgins JP, Altman DG, Gotzsche PC, Juni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011; 343: d5928.	Assessment of RCTs
Campos TF, Beckenkamp PR, Moseley AM. Usage evaluation of a resource to support evidence-based physiotherapy: the physiotherapy evidence database (PEDro). Physiotherapy 2013; 99(3): 252–57.	Assessment of RCTs

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Jadad AR, Moore RA, Carroll D, Jenkinson C, Reynolds DJ, Gavaghan DJ, et al. Assessing the quality of reports of randomized clinical trials: is blinding necessary? Controlled Clinical Trials 1996; 17(1): 1–12	Assessment of RCTs
Verhagen AP, de Vet HC, de Bie RA, Kessels AG, BoersM, Bouter LM, et al. The Delphi list: a criteria list for quality assessment of randomized clinical trials for conducting systematic reviews developed by Delphi consensus. Journal of Clinical Epidemiology 1998; 51(12): 1235–41.	Assessment of RCTs
Moga C, Guo B, Schopflocher D, Harstall C. Development of a Quality Appraisal Tool for Case Series Studies using a Modified Delphi Technique. 2012.	Assessment of case series studies
Stroke Therapy Academic Industry R. Recommendations for standards regarding preclinical neuroprotective and restorative drug development. Stroke 1999; 30(12): 2752–58.	Assessment of animal studies
Fisher M, Feuerstein G, Howells DW, Hurn PD, Kent TA, Savitz SI, et al. Update of the stroke therapy academic industry roundtable preclinical recommendations. Stroke 2009; 40(6): 2244–50.	Assessment of animal studies
Macleod MR, O'Collins T, Howells DW, Donnan GA. Pooling of animal experimental data reveals influence of study design and publication bias. Stroke 2004; 35(5): 1203–08.	Assessment of animal studies
Shea BJ, Grimshaw JM, Wells GA, Boers M, Andersson N, Hamel C, et al. Development of AMSTAR: a measurement tool to assess the methodological quality of systematic reviews. BMC Medical Research Methodology 2007; 7: 10.	Assessment of systematic reviews
Collaboration A. Development and validation of an international appraisal instrument for assessing the quality of clinical practice guidelines: the AGREE project. Quality & Safety in Health Care 2003; 12(1): 18–23	Assessment of clinical practice guidelines
Brouwers MC, Kho ME, Browman GP, Burgers JS, Cluzeau F, Feder G, et al. AGREE II: advancing guideline development, reporting and evaluation in health care. Canadian Medical Association Journal 2010; 182(18): E839–E842.	Assessment of clinical practice guidelines
Kirkham JJ, Dwan KM, Altman DG, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. BMJ 2010;340:c365.	Assessment of RCTs
Meader N, King K, Llewellyn A, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. Systematic reviews 2014;3(1):82.	Assessment of systematic reviews
Dwan K, Gamble C, Kolamunnage-Dona R, et al. Assessing the potential for outcome reporting bias in a review: a tutorial. Trials 2010;11:52.	Assessment of RCTs
Stewart GB, Higgins JP, Schunemann H, et al. The use of Bayesian networks to assess the quality of evidence from research synthesis: 1. PLoS One 2015;10(3):e0114497.	Assessment of systematic reviews
Salanti G, Giovane CD, Chaimani A, et al. Evaluating the quality of evidence from a network meta-analysis. PLoS One 2014;9(7):e99682.	Assessment for network meta-analyses
Higgins JP, Del Giovane C, Chaimani A, et al. Evaluating the quality of evidence from a network meta-analysis. Value Health 2014;17(7):A324.	Assessment for network meta-analyses
WP5, EMA, EUnetHTA Joint Action 3 (2016 - 2020), REQueST® tool. Available at: https://eunethta.eu/wp-content/uploads/2019/10/EUnetHTAJA3_Vision_paper-on-REQueST-tool.pdf	Assessment specific on the quality of registries

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Figure S3. Assessment tools that included or not at least one question/item to address subdomains of methods for selecting study participants (Domain 1), stratified by study design.

	Domain 1 - Methods for selecting participants (SELECTION BIAS)				
Tools for Case-Control Studies only	Sampling strategies to correct selection bias	Inclusion and exclusion criteria of target population	Depletion of susceptibles	Generalizability of target population	Others^
RAMboMAN (GATE-EPIQ) 2019	×	✓	×	✓	✓
CASP 2018	✓	✓	×	✓	✓
SURE 2018	✓	✓	×	×	✓
JBI 2017	×	×	×	×	✓
NIH–NHLBI 2014	✓	✓	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	✓	✓
SIGN Version 3.0 2012	×	✓	×	✓	✓
STROBE statement V4 2007	✓	✓	×	×	✓
Durant 1994	×	×	×	✓	✓
Gyorkos 1994	×	×	×	×	✓
Lichtenstein 1987	✓	✓	×	×	✓
Horwitz 1979	✓	✓	×	×	✓
Tools for Cohort Studies only					
RAMboMAN (GATE-EPIQ) 2019	×	✓	×	✓	✓
CASP 2018	✓	×	×	✓	✓
SURE 2018	✓	✓	×	×	✓
JBI 2017	×	✓	×	×	✓
NIH–NHLBI 2014	×	✓	×	✓	✓
HEB Wales 2014	×	×	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	✓	✓
SIGN Version 3.0 2012	×	×	×	×	✓
STROBE statement V4 2007	✓	✓	×	×	✓
Hadorn 1996	✓	✓	×	✓	✓
Gyorkos 1994	×	×	×	×	✓
Tools for multiple NRS					
RELEVANT 2019	×	✓	×	✓	✓
MMAT 2018	×	✓	×	✓	×
ROBINS-I 2016	×	✓	✓	×	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	×	×	✓
GRACE 2014 (#)	×	×	×	×	×
RoBANS 2013	×	×	×	×	✓
RTI 2013	✓	✓	×	×	✓
Montreal 2011	✓	×	×	✓	×
EPHPP 2004	×	×	×	✓	✓
TREND statement 2004	✓	✓	×	×	✓
Margetts 2002	✓	✓	×	×	✓
Zaza 2000	✓	✓	×	✓	✓
Downs-Black 1998	×	✓	×	✓	✓
Elwood 1998	×	✓	×	×	✓
London 1996	×	×	×	✓	✓
Avis 1994	✓	×	×	×	✓
Levine 1994	×	×	×	×	✓
Cho 1994 (#)	×	✓	×	×	✓
COEH 1991	✓	✓	×	✓	✓
Fowkes-Fulton 1991	✓	✓	×	✓	✓
Gardner 1986	×	×	×	×	✓

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
^ "Others" refers to items not included in our evaluation framework but included in the reviewed tools to investigate selection bias
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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Figure S4. Assessment tools that included or not at least one question/item to address subdomains for measurement of exposure, outcomes, covariate variables and follow-up (Domain 2), stratified by study design.

	Domain 2 - Measurement of exposure, outcomes, covariate variables and follow-up.			
Tools for Case-Control Studies only	Exposure	Outcome	Covariate variables	Follow-up
RAMboMAN (GATE-EPIQ) 2019	✓	✓	×	✓
CASP 2018	✓	✓	✓	×
SURE 2018	✓	✓	×	×
JBI 2017	✓	✓	×	✓
NIH–NHLBI 2014	✓	✓	×	×
Newcastle-Ottawa 2013	✓	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×	×
STROBE statement V4 2007	✓	✓	✓	✓
Durant 1994	×	×	✓	×
Gyorkos 1994	✓	✓	×	×
Lichtenstein 1987	✓	✓	×	×
Horwitz 1979	✓	✓	×	×
Tools for Cohort Studies only				
RAMboMAN (GATE-EPIQ) 2019	✓	✓	×	✓
CASP 2018	✓	✓	✓	✓
SURE 2018	✓	✓	×	✓
JBI 2017	✓	✓	×	✓
NIH–NHLBI 2014	✓	✓	×	✓
HEB Wales 2014	✓	✓	✓	✓
Newcastle-Ottawa 2013	✓	✓	✓	✓
SIGN Version 3.0 2012	✓	✓	×	×
STROBE statement V4 2007	✓	✓	✓	✓
Hadorn 1996	✓	✓	×	×
Gyorkos 1994	✓	✓	×	✓
Tools for multiple NRS				
RELEVANT 2019	✓	✓	×	×
MMAT 2018	✓	✓	×	×
ROBINS-I 2016	✓	✓	✓	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	×	✓
GRACE 2014 (#)	✓	✓	✓	×
RoBANS 2013	✓	✓	×	×
RTI 2013	✓	✓	✓	✓
Montreal 2011	×	✓	✓	×
EPHPP 2004	✓	✓	×	×
TREND statement 2004	✓	✓	×	✓
Margetts 2002	✓	✓	×	×
Zaza 2000	✓	✓	×	×
Downs-Black 1998	✓	✓	×	×
Elwood 1998	✓	✓	×	×
London 1996	✓	✓	×	×
Avis 1994	✓	✓	×	×
Levine 1994	✓	✓	×	✓
Cho 1994 (#)	×	×	×	×
COEH 1991	✓	×	×	×
Fowkes-Fulton 1991	✓	✓	×	×
Gardner 1986	×	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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Figure S5. Assessment tools that included or not at least one question/item to address subdomains to detect design-specific sources of biases (Domain 3), stratified by study design.

	Domain 3 - Design-specific sources of bias						
Tools for Case-Control Studies only	new user design	active comparator design	immortal time bias	detection or surveillance bias	loss to follow-up bias	Non-contemporaneous comparator bias	Reverse causation
RAMboMAN (GATE-EPIQ) 2019	×	×	×	×	×	×	×
CASP 2018	×	×	×	×	×	×	×
SURE 2018	×	×	×	×	×	×	×
JBI 2017	×	×	×	×	×	×	×
NIH–NHLBI 2014	×	×	×	×	×	✓	✓
Newcastle-Ottawa 2013	×	×	×	×	✓	×	×
SIGN Version 3.0 2012	×	×	×	×	×	×	×
STROBE statement V4 2007	×	×	×	×	×	×	×
Durant 1994	×	×	×	✓	×	×	×
Gyorkos 1994	×	×	×	×	×	×	×
Lichtenstein 1987	×	×	×	×	×	×	×
Horwitz 1979	×	×	×	✓	×	×	×
Tools for Cohort Studies only							
RAMboMAN (GATE-EPIQ) 2019	×	×	×	×	✓	×	✓
CASP 2018	×	×	×	×	×	×	×
SURE 2018	×	×	×	×	✓	×	×
JBI 2017	×	×	×	×	✓	×	✓
NIH–NHLBI 2014	×	×	×	×	✓	×	✓
HEB Wales 2014	×	×	×	×	✓	×	×
Newcastle-Ottawa 2013	×	×	×	×	✓	×	✓
SIGN Version 3.0 2012	×	×	×	✓	✓	×	✓
STROBE statement V4 2007	×	×	×	×	✓	×	×
Hadorn 1996	×	×	×	×	✓	×	×
Gyorkos 1994	×	×	×	×	×	×	×
Tools for multiple NRS							
RELEVANT 2019	×	×	×	×	×	×	×
MMAT 2018	×	×	×	×	✓	×	×
ROBINS-I 2016	✓	×	✓	×	✓	✓	✓
ISPOR-AMCP-NPC 2014 (#)	×	×	×	×	×	✓	×
GRACE 2014 (#)	✓	×	✓	×	×	✓	×
RoBANS 2013	×	×	×	✓	✓	✓	×
RTI 2013	×	×	×	×	✓	×	×
Montreal 2011	×	×	×	×	✓	×	×
EPHPP 2004	×	×	×	×	✓	×	×
TREND statement 2004	×	×	×	×	×	×	×
Margetts 2002	×	×	×	×	×	×	×
Zaza 2000	×	×	×	×	✓	×	×
Downs-Black 1998	×	×	×	×	✓	✓	×
Elwood 1998	×	×	×	×	×	×	✓
London 1996	×	×	✓	×	✓	×	×
Avis 1994	×	×	×	×	✓	×	×
Levine 1994	×	×	×	×	×	×	×
Cho 1994 (#)	×	×	×	×	✓	×	×
COEH 1991	×	×	×	×	×	×	×
Fowkes-Fulton 1991	×	×	×	×	✓	×	×
Gardner 1986	×	×	×	×	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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Figure S6. Assessment tools that included or not at least one question/item to address confounding (Domain 4), stratified by study design.

	Domain 4 - Confounding		
Tools for Case-Control Studies only	study design used to minimize the confounding	confounders measured and included in the analyses	potential unmeasured confounding addressed in the analyses
RAMboMAN (GATE-EPIQ) 2019	✓	×	×
CASP 2018	✓	✓	×
SURE 2018	×	✓	×
JBI 2017	✓	✓	×
NIH–NHLBI 2014	×	✓	×
Newcastle-Ottawa 2013	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×
STROBE statement V4 2007	×	✓	×
Durant 1994	✓	✓	✓
Gyorkos 1994	×	✓	×
Lichtenstein 1987	✓	✓	×
Horwitz 1979	×	×	×
Tools for Cohort Studies only			
RAMboMAN (GATE-EPIQ) 2019	✓	×	×
CASP 2018	✓	✓	×
SURE 2018	×	✓	×
JBI 2017	✓	✓	×
NIH–NHLBI 2014	×	✓	×
HEB Wales 2014	×	✓	✓
Newcastle-Ottawa 2013	✓	✓	×
SIGN Version 3.0 2012	✓	✓	×
STROBE statement V4 2007	×	✓	×
Hadorn 1996	✓	✓	×
Gyorkos 1994	×	✓	×
Tools for multiple NRS			
RELEVANT 2019	✓	✓	×
MMAT 2018	✓	✓	×
ROBINS-I 2016	×	✓	✓
ISPOR-AMCP-NPC 2014 (#)	✓	✓	✓
GRACE 2014 (#)	✓	✓	×
RoBANS 2013	✓	✓	×
RTI 2013	✓	✓	×
Montreal 2011	×	✓	×
EPHPP 2004	✓	✓	×
TREND statement 2004	✓	✓	×
Margetts 2002	×	×	×
Zaza 2000	✓	✓	✓
Downs-Black 1998	×	✓	×
Elwood 1998	✓	✓	×
London 1996	✓	✓	×
Avis 1994	×	✓	×
Levine 1994	×	×	×
Cho 1994 (#)	✓	✓	×
COEH 1991	×	✓	×
Fowkes-Fulton 1991	✓	✓	×
Gardner 1986	×	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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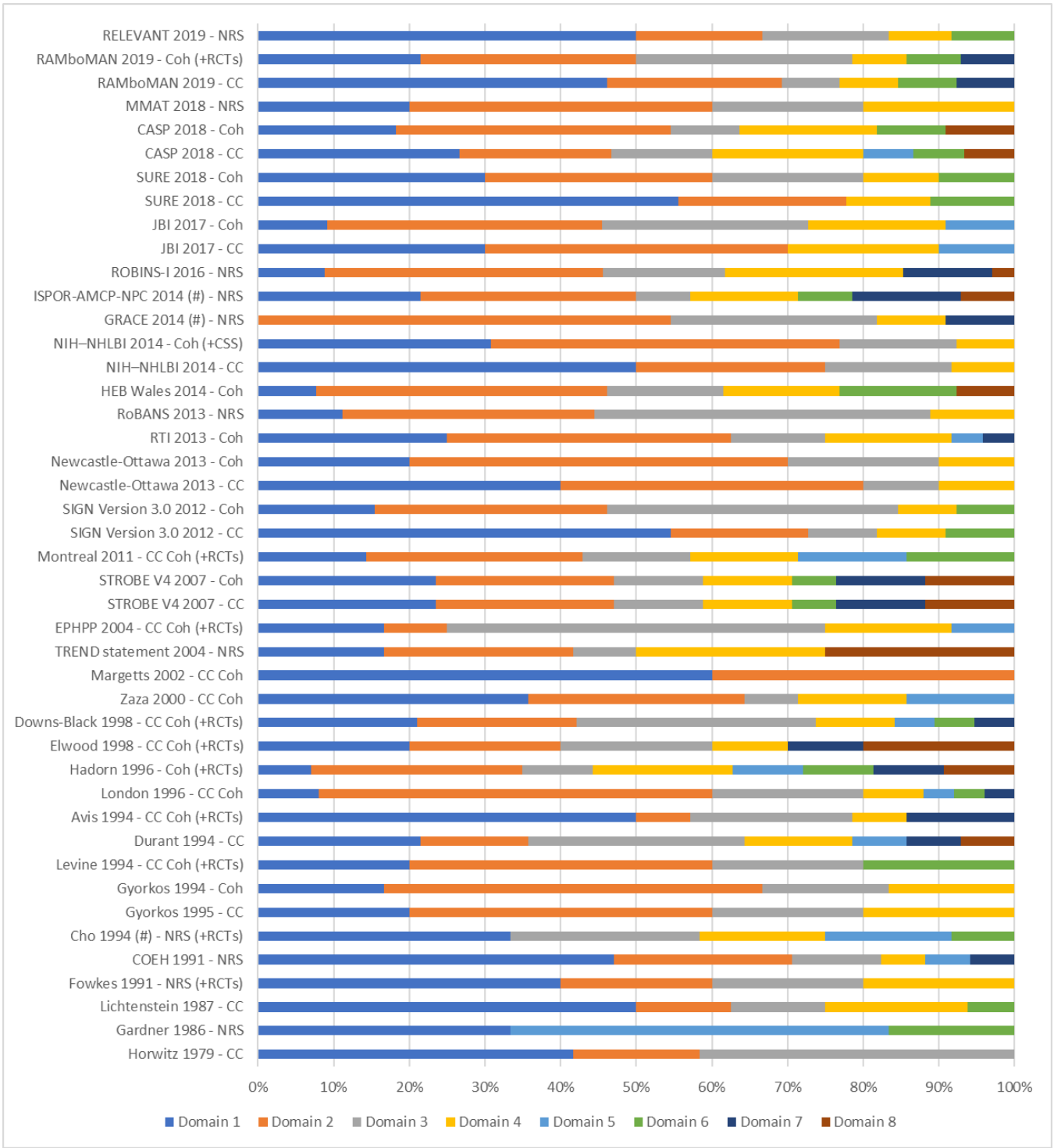
Figure S7. Assessment tools that included or not at least one question/item to address lack of appropriateness of statistical analyses (Domain 5), methods for assessing statistical uncertainty in the findings (Domain 6), methods for assessing internal validity (Domain 7), methods for assessing external validity (Domain 8), stratified by study design.

	Domain 5 - Lack of appropriate statistical analyses	**Domain 6 - Statistical uncertainty**	**Domain 7 - Internal validity**	**Domain 8 - External validity**
Tools for Case-Control Studies only	(e.g., overadjustment for intermidiates, incorrect outcome model specification)	(e.g., confidence intervals reported for each analysis)	(e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases)	(e.g. post-hoc subgroup analysis, comparison with other populations etc.)
RAMboMAN (GATE-EPIQ) 2019	×	✓	✓	×
CASP 2018	✓	✓	×	✓
SURE 2018	×	✓	×	×
JBI 2017	✓	×	×	×
NIH–NHLBI 2014	×	×	×	×
Newcastle-Ottawa 2013	×	×	×	×
SIGN Version 3.0 2012	×	✓	×	×
STROBE statement V4 2007	×	✓	✓	✓
Durant 1994	✓	×	✓	✓
Gyorkos 1994	×	×	×	×
Lichtenstein 1987	×	✓	×	×
Horwitz 1979	×	×	×	×
Tools for Cohort Studies only				
RAMboMAN (GATE-EPIQ) 2019	×	✓	✓	×
CASP 2018	×	✓	×	✓
SURE 2018	×	✓	×	×
JBI 2017	✓	×	×	×
NIH–NHLBI 2014	×	×	×	×
HEB Wales 2014	×	✓	×	✓
Newcastle-Ottawa 2013	×	×	×	×
SIGN Version 3.0 2012	×	✓	×	×
STROBE statement V4 2007	×	✓	✓	✓
Hadorn 1996	✓	✓	✓	✓
Gyorkos 1994	×	×	×	×
Tools for multiple NRS				
RELEVANT 2019	×	✓	×	×
MMAT 2018	×	×	×	×
ROBINS-I 2016	✓	×	✓	✓
ISPOR-AMCP-NPC 2014 (#)	×	✓	✓	✓
GRACE 2014 (#)	×	×	✓	×
RoBANS 2013	×	×	×	×
RTI 2013	✓	×	✓	×
Montreal 2011	✓	✓	×	×
EPHPP 2004	✓	×	×	×
TREND statement 2004	×	×	×	✓
Margetts 2002	×	×	×	×
Zaza 2000	✓	×	×	×
Downs-Black 1998	✓	✓	✓	×
Elwood 1998	×	×	✓	✓
London 1996	✓	✓	✓	×
Avis 1994	×	×	✓	×
Levine 1994	×	✓	×	×
Cho 1994 (#)	✓	✓	×	×
COEH 1991	✓	×	✓	×
Fowkes-Fulton 1991	×	×	×	×
Gardner 1986	✓	✓	×	×

Legend: ✓ = at least one item/question addressing the subdomain; × = not items/questions addressing the subdomain.
(#) tool developed to assess NRS on the comparative safety and effectiveness of medications.

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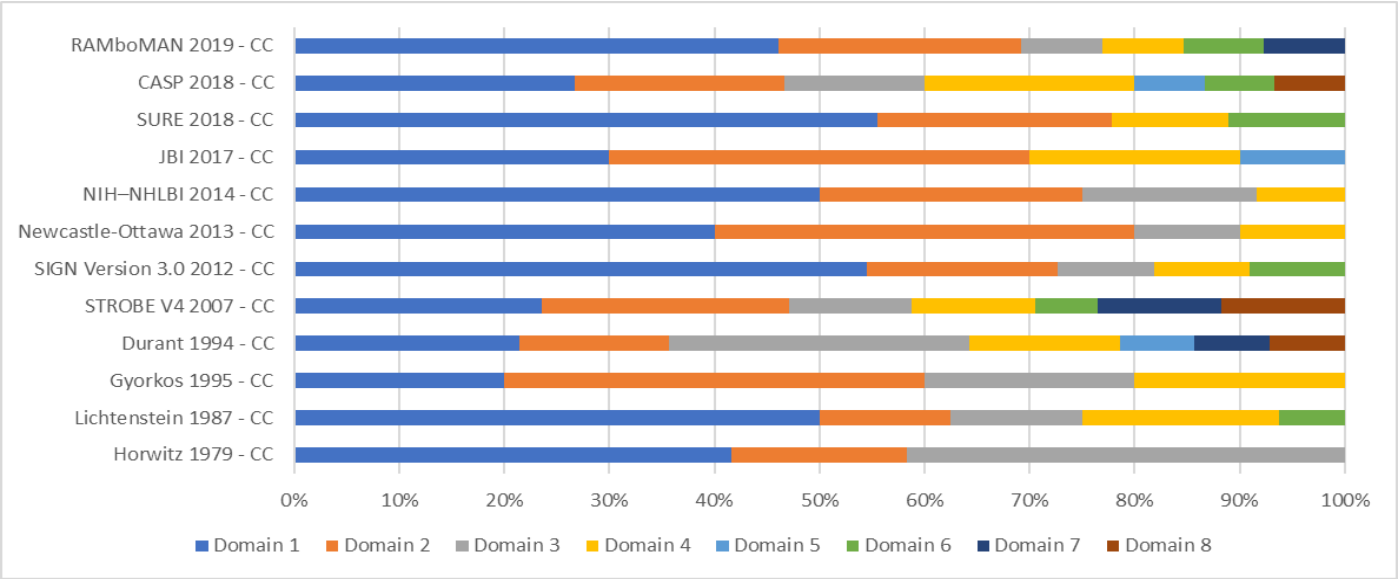
Figure S8. Proportion of domains covered by each reviewed tool



Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

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Figure S9. Proportion of domains covered by each reviewed tool of case-control studies

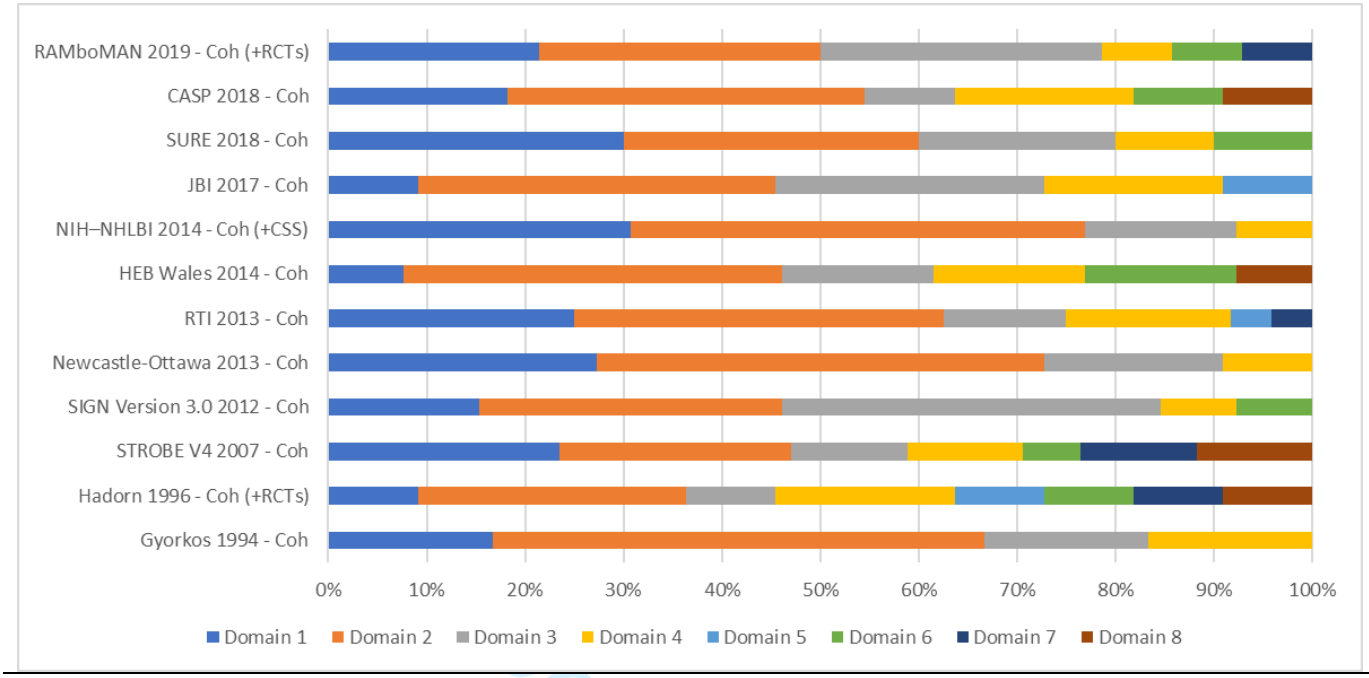


Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

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Figure S10. Proportion of domains covered by each reviewed tool of cohort studies

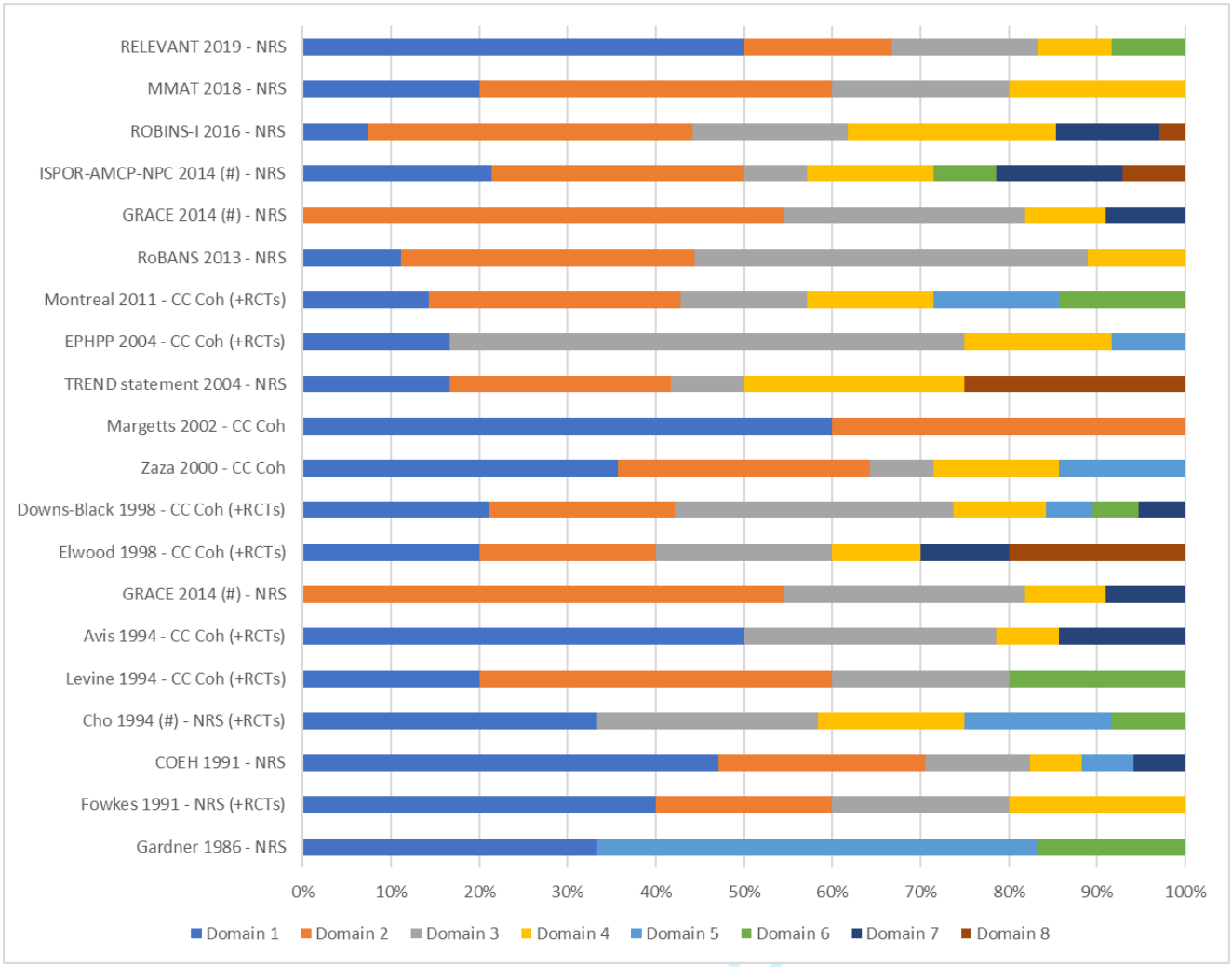


Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

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Figure S11. Proportion of domains covered by each reviewed tool of multiple study designs (NRS)



Legend: Domain 1 = Methods for selecting participants; Domain 2 = Definition and measurement of exposure, outcomes, covariates and follow-up; Domain 3 = Design-specific sources of bias; Domain 4 = Confounding; Domain 5 = Lack of appropriateness of statistical analyses; Domain 6 = Methods for assessing statistical uncertainty in the findings; Domain 7 = Methods for assessing internal validity; Domain 8 = Methods for assessing external validity.

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Reference

Hasson F, Keeney S, McKenna H. Research guidelines for the Delphi survey technique. J Adv Nurs. 2000;32(4):1008-1015.

Sørensen HT, Lash TL, Rothman KJ. Beyond randomized controlled trials: a critical comparison of trials with nonrandomized studies. Hepatology. 2006;44(5):1075-1082

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Supplement file 2

Survey to the experts of the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research (CER) - Special Interest Group: FRAMEWORK TO EVALUATE ASSESSMENT TOOLS

Dear Member,

As an expert contributor to the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research (CER) Special Interest Group, we would like to invite you to take part in a short survey aimed at identifying key quality domains that:
(i) should be considered when assessing the overall quality of a nonrandomized study (NRS) in pharmacoepidemiology, and;
(ii) are not covered by existing appraisal tools for assessing the quality of NRSs.

Your answers will be used to inform the research project on an evaluation of existing tools to assess the methodological quality of pharmacoepidemiological NRSs - specifically cohort and case-control studies.

The survey will take approximately 5 minutes to complete and consists of two sections:
A. Demographic and professional information
B. Evaluation of key quality domains

In SECTION A, you will be asked to provide your demographic and professional information.

SECTION B lists key domains in assessing the overall quality of NRSs in pharmacoepidemiology.
Each key domain is presented as an item that can be part of a tool for evaluating the methodological quality of pharmacoepidemiological NRSs (cohort and case control designs). You are asked to evaluate the importance of each item, rating it from 1 (not important) to 5 (extremely important). At the end of section B, you will have the opportunity to report additional items that were not presented in the survey, are key for evaluating the methodological quality of pharmacoepidemiological NRSs and are not included in current appraisal tools.

Your answers will be kept strictly confidential and will be used only for the purposes of this research project. The results will be reported in aggregate form only and will be part of a publication in a peer-reviewed journal.

Should you have any questions on the survey or the project, please contact Elvira D'Andrea at edandrea@bwh.harvard.edu or Lydia Vinals at lydia.vinals@evidera.com.

We thank you for your collaboration.

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Thomas Debray (t.debray@umcutrecht.nl)
Research Project coordinators.

Elisabetta Patorno (epatorno@bwh.harvard.edu)
Chair, ISPE CER Special Interest Group.

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Section A – Demographic and professional info

Institution or Organization:

– Academia

– Industry

– Governmental Agencies

– Other (specify) _____

Professional position _____

Country _____

Section B - Evaluation of key quality domains

Please rate the following items from 1 (not important) to 5 (extremely important). Your judgment should be based on the importance that the key domains described by each item should have in tools assessing the overall methodological quality of a pharmacoepidemiologic NRS.

How important is it for a tool to assess:

1. Methods for selecting participants (selection bias)

Not important	1	2	3	4	5	Extremely important

Specifically:

1.1 Sampling strategies to correct selection bias

Not important	1	2	3	4	5	Extremely important

2.1 Inclusion and exclusion criteria of target population clearly defined

Not important	1	2	3	4	5	Extremely important

2.2 Depletion of susceptibles

Not important	1	2	3	4	5	Extremely important

2.3 Methods of selecting participants*

Not important	1	2	3	4	5	Extremely important

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2.4 Other key source of selection bias not included above _____

*question added to the questionnaire of the second round of the Delphi survey based on the answers provided on the first round.

2. Definition and measurement of exposure, outcomes, covariate variables and follow-up.

Not important	1	2	3	4	5	Extremely important

Specifically:

2.1 Definition and measurement of exposure

Not important	1	2	3	4	5	Extremely important

2.2 Definition and measurement of outcomes

Not important	1	2	3	4	5	Extremely important

2.3 Definition and measurement of covariate variables

Not important	1	2	3	4	5	Extremely important

2.4 Definition and measurement of follow-up

Not important	1	2	3	4	5	Extremely important

3. Design-specific sources of bias:

Not important	1	2	3	4	5	Extremely important

Specifically:

3.1 lack of use of new user design:

Not important	1	2	3	4	5	Extremely important

3.2 lack of use of active comparator:

Not important	1	2	3	4	5	Extremely important

3.3 Time-related biases (immortal time bias or window-time bias):

Not important	1	2	3	4	5	Extremely important

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3.4 Detection or surveillance bias:

Not important	1	2	3	4	5	Extremely important

3.5 Loss to follow-up bias:

Not important	1	2	3	4	5	Extremely important

3.6 Non-contemporaneous comparator bias:

Not important	1	2	3	4	5	Extremely important

3.7 Reverse causation:

Not important	1	2	3	4	5	Extremely important

3.8 Recall bias[#]:

Not important	1	2	3	4	5	Extremely important

3.10 Other critical design-specific sources of bias not mentioned above _____

[#] question removed to the questionnaire of the second round of the Delphi survey based on the answers provided on the first round.

4. Confounding:

Not important	1	2	3	4	5	Extremely important

Specifically:

4.1 Study design used to minimize the confounding:

Not important	1	2	3	4	5	Extremely important

4.2 Confounders measured and included in the analyses

Not important	1	2	3	4	5	Extremely important

4.3 Potential unmeasured confounding (addressed in the analysis through proxy analysis, IV analysis, etc.)

Not important	1	2	3	4	5	Extremely important

4.4 Other critical elements for confounding not mentioned above _____

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5. Lack of appropriateness of statistical analyses (with specific mention of adjustment for causal intermediaries, and/or incorrect outcome model specification)

Not important	1	2	3	4	5	Extremely important

6. Methods for assessing statistical uncertainty in the findings (e.g., confidence intervals reported for each analysis)

Not important	1	2	3	4	5	Extremely important

7. Methods for assessing internal validity (e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases)

Not important	1	2	3	4	5	Extremely important

8. Methods for assessing external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.)

Not important	1	2	3	4	5	Extremely important

9. Any other key considerations not listed above _____

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GLOSSARY

SELECTION BIAS defined as a bias due to a sample selection that does not accurately reflect the target population.

DEPLETION OF SUSCEPTIBLES is defined as a phenomenon where the initiation of exposure to a drug is associated with an early increase in the incidence rate of an acute event, followed by a decrease in this incidence rate with longer duration of exposure.

NEW USER DESIGN, defined as a study design that starts following patients at the time they initiate a new drug (also known as incident user design).

ACTIVE COMPARATOR DESIGN, defined as a study design that compares the effect of the drug of interest with another drug used in clinical practice instead of non-use.

IMMORTAL TIME BIAS, defined as a bias that derives from including a period of follow-up during which, by design, outcomes cannot occur.

WINDOW-TIME BIAS, in the context of a case-control study nested in a cohort, defined as a bias that derives from the use of time-windows of different lengths between cases and controls to define time-dependent exposures.

DETECTION OR SURVEILLANCE BIAS, defined as a bias that derives from systematic differences in outcomes assessment among groups being compared.

NON-CONTEMPORANEOUS COMPARATOR BIAS, defined as a bias generated by differences in the timing of selection of comparator group(s) within a study influence exposures and outcomes resulting in biased estimates.

REVERSE CAUSATION (or reverse causality), defined as a direction of cause-and-effect contrary to a common presumption, or a two-way causal relationship between exposure and outcome.

RECALL BIAS, defined as a bias that occurs when participants do not remember previous events or experiences accurately or omit details (not for claims-based studies).

LOSS TO FOLLOW-UP BIAS, defined as a bias that occurs when there is difference in retention during the follow-up period after enrollment that are related to exposure status and outcome. It can be considered as part of selection bias.

CONFOUNDING, defined as a mixing of effects that arises when patients with different baseline risks are compared – the resulting effect measure is a mix of drug effects and risk factor effects.

ADJUSTMENT FOR CAUSAL INTERMEDIARIES, defined as adjustment for an intermediate variable (or a descending proxy for an intermediate variable) on a causal path from exposure to outcome.

INCORRECT OUTCOME MODEL SPECIFICATION, for omission of a relevant variable, inclusion of an unnecessary variable, adopting the wrong functional form, incorrect specification of the error term, uncertainty about what the true model is, reciprocal causation, etc.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	# 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	# 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	# 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	# 4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	# 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	# 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	# 5 and Suppl. 1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	# 5 and Suppl. 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	# 6-7 and Suppl. 1
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	# 5-6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	# 5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	# 5-6 (framework)
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	# 6

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	---

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	---
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	# 6 and Suppl. 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	# 7-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	# 7-9
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Suppl. 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	---
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Suppl. 1
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	# 7-9
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	# 9-10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	# 12-13
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	# 13
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	# 13-14

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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